Principal Bundles and Applications

Bowen Liu[1]

[1]liubw22@mails.tsinghua.edu.cn

Contents

1. To the Reader

1.1. **About this lecture.** These are lecture notes from a seminar that I organized in Spring 2023 to study the basic theory of principal bundles and their applications. These notes are divided into the following two parts:

(1) In the **first part:** First, we are mainly concerned with the basic theory of principal bundles, such as what a principal bundle is, associated fiber bundles, and reductions of principal bundles. Then we introduce the heart of these notes: local computations for connections and curvature on principal bundles. At the end of this part, we also show how to obtain a connection on an associated vector bundle from one on a principal bundle, and we discuss the classical Riemann-Hilbert correspondence.
(2) In the **second part** we focus on Chern-Weil theory, which allows us to construct characteristic classes from curvature and invariant polynomials. We also give a brief introduction to classifying spaces for principal bundles and their relation to characteristic classes.

1.2. **Acknowledgement.** I would like to express my heartfelt gratitude to Qiliang Luo, Shan Li, Shihao Wang, Yifei Cai, Yuxuan Li, and Zhitong Chen for their attendance and discussions. I am particularly indebted to Zhiyao Xiong for his enthusiasm, without which I could not have learned so much.

1.3. **Some notations.**

1.3.1. *On base manifold.*

(1) M is used to denote a smooth manifold, and $x \in M$ denotes a point of M.
(2) TM and Ω_M^k are used to denote the tangent bundle and the bundle of k-forms over M, respectively.
(3) $\Omega_M^k(E)$ is used to denote the bundle of k-forms over M with values in E.
(4) v is used to denote a vector in a tangent space.
(5) X is used to denote a vector field on M, and X_x denotes the value of X at the point $x \in M$.
(6) α is used to denote a k-form on M, and α_x denotes the value of α at the point $x \in M$.
(7) For a vector bundle E over M, $C^\infty(E,M)$ is used to denote its sections.

1.3.2. *On principal bundle.*

(1) G is used to denote a Lie group, with Lie algebra $\mathfrak{g}$.
(2) $\pi : P \to M$ is used to denote a principal G-bundle over M, and $p \in P$ denotes its point.
(3) $\widetilde{X}$ is used to denote a vector field on the principal bundle P; similarly for $\widetilde{\alpha}$ and $\widetilde{v}$.
(4) ω is used to denote a connection 1-form on P, with curvature 2-form Ω.

Part 1. **Principal bundle and its geometry**

2. PRINCIPAL BUNDLE

2.1. **A glimpse of fiber bundle.**

Definition 2.1.1 (fiber bundle)**.** Let F, E, B be topological spaces. A fiber bundle with fiber F over B is a surjective map $\pi \colon E \to B$ such that for any $b \in B$, there exists an open neighborhood $U \ni b$ and a homeomorphism φ such that the following diagram commutes

$$\begin{tikzcd} \pi^{-1}(U) \arrow[rr, "\varphi"] \arrow[dr, "\pi"'] & & U \times F \arrow[dl, "\pi_1"] \\ & U & \end{tikzcd}$$

We write $F \to E \xrightarrow{\pi} B$ or (E, B, π, F) for this fiber bundle, and

(1) B is called the base space.
(2) $E_x = \pi^{-1}(x)$ is called the fiber of E at x.
(3) (U, φ) is called a local trivialization at b, and we use $E|_U$ to denote $\pi^{-1}(U)$.

Example 2.1.1 (trivial bundle)**.** Consider $E = B \times F$, with $\pi \colon E \to B$ given by the projection onto the first summand.

Example 2.1.2. Consider $E = S^n$ and $B = \mathbb{RP}^n$. Then the natural map $\pi \colon E \to B$ is a fiber bundle with fiber $\mathbb{Z}/2\mathbb{Z}$. It is clear that this fiber bundle is not trivial, since S^n is connected.

Example 2.1.3 (Hopf fibration)**.** Recall that

$$\mathbb{CP}^n = \{\text{the set of all complex lines through the origin in } \mathbb{C}^{n+1}\}$$

Consider the canonical open cover $\{U_i\}$ of $\mathbb{CP}^n$, that is

$$U_i = \{[z_0 : \ldots : z_n] \mid z_i \neq 0\}$$

Now view $S^{2n+1} \subseteq \mathbb{R}^{2n+2} = \mathbb{C}^{n+1}$ as the set of all $(z_0, \ldots, z_n) \in \mathbb{C}^{n+1}$ with $|z_0|^2 + \cdots + |z_n|^2 = 1$. Then the projection map $\pi \colon \mathbb{C}^{n+1} - \{0\} \to \mathbb{CP}^n$ restricts to a surjective smooth map

$$\pi \colon S^{2n+1} \to \mathbb{CP}^n$$

We claim that it is a fiber bundle with fiber S^1. Indeed, by definition we have

$$\pi^{-1}(U_i) = \{(z_0, \ldots, z_n) \in S^{2n+1} \mid z_i \neq 0\}$$

and a local trivialization map can be taken to be

$$\begin{aligned} \varphi_i \colon \pi^{-1}(U_i) &\to U_i \times S^1 \\ z &\mapsto ([z_0 : \ldots : z_n], \frac{z_i}{|z_i|}) \end{aligned}$$

It is also nontrivial, as can be seen by considering the fundamental groups.

Example 2.1.4. A covering space is a fiber bundle with a discrete set as its fiber.

2.2. **Principal bundle.**

2.2.1. *Definitions.* Briefly speaking, given a Lie group G and a smooth manifold M, a principal G-bundle P is a fiber bundle with fiber G equipped with a suitable smooth right G-action on it. By a smooth right G-action we mean a smooth map

$$\begin{aligned} P \times G &\to P \\ (p,g) &\mapsto pg \end{aligned}$$

Definition 2.2.1 (principal G-bundle)**.** A principal G-bundle is a surjective smooth map $\pi\colon P \to M$ between smooth manifolds such that:

(1) There is a smooth right G-action on P.
(2) For all $x \in M$, $\pi^{-1}(x)$ is a G-orbit.
(3) For all $x \in M$, there exists an open subset U_α containing x and a G-equivariant diffeomorphism φ_α, which is called a local trivialization, such that the following diagram commutes

$$\begin{tikzcd} \pi^{-1}(U_\alpha) \arrow[rr, "\varphi_\alpha"] \arrow[dr, "\pi"'] & & U_\alpha \times G \arrow[dl, "\pi_{U_\alpha}"] \\ & U_\alpha & \end{tikzcd}$$

Notation 2.2.1. $\mathcal{P}_G M$ is used to denote the set of all principal G-bundles over M up to isomorphism.

Remark 2.2.1. If we write $\varphi_\alpha(p) = (\pi(p), g_\alpha(p))$, then φ_α is G-equivariant if and only if $g_\alpha(pg) = g_\alpha(p)g$ for any $g \in G$.

Proposition 2.2.1. Let P be a principal G-bundle. Then G acts freely on P and transitively on each fiber.

Proof. This is clear from a local trivialization. □

Example 2.2.1. $S^n \to \mathbb{RP}^n$ is a $\mathbb{Z}/2\mathbb{Z}$-principal bundle, where $\mathbb{Z}/2\mathbb{Z}$ acts on S^n via $x \mapsto -x$.

Example 2.2.2. $S^{2n+1} \to \mathbb{CP}^n$ is a $\mathrm{U}(1)$-principal bundle, where $\mathrm{U}(1)$ acts on S^{2n+1} via $(z_0, z_1, \ldots, z_n) \mapsto (z_0 e^{i\theta}, z_1 e^{i\theta}, \ldots, z_n e^{i\theta})$.

Definition 2.2.2 (morphism between principal G-bundles)**.** For two principal G-bundles $(P, M, \pi), (P', M, \pi')$, a morphism between them is a G-equivariant smooth map $\varphi\colon P \to P'$ making the following diagram commute

$$\begin{tikzcd} P \arrow[rr, "\varphi"] \arrow[dr, "\pi"'] & & P' \arrow[dl, "\pi'"] \\ & M & \end{tikzcd}$$

Proposition 2.2.2. A morphism $\varphi\colon P \to P'$ between principal G-bundles over M is an isomorphism.

Proof. All information is encoded in the G-equivariance of φ and properties of principal G-bundles:

(1) φ is injective: For any $p_1, p_2 \in P$, if $\varphi(p_1) = \varphi(p_2)$, then p_1, p_2 lie in the same fiber, since the above diagram commutes. If $p_1 = p_2 g$ for $g \in G$, then $\varphi(p_1) = \varphi(p_2)g$, which implies $g = e$, since G acts on P' freely, that is $p_1 = p_2$.
(2) φ is surjective: For any $p' \in P'$, if $\pi'(p') = x$, then $p' \in P'_x$. So choose an arbitrary element $p \in P_x$, there must be some $g \in G$ such that $\varphi(pg) = p'$, since P'_x is a G-orbit and φ is G-equivariant.

In local trivializations, φ has the form $(x,h) \mapsto (x, a(x)h)$ for a smooth map $a\colon U \to G$. Its inverse is $(x,h) \mapsto (x, a(x)^{-1}h)$, so the bijection is a diffeomorphism. □

Definition 2.2.3 (trivial principal bundle)**.** A principal G-bundle P is called a trivial principal bundle, if there exists a principal G-bundle isomorphism $\varphi\colon P \to M \times G$.

2.2.2. *Structure group.* Let $\{U_\alpha, \varphi_\alpha\}$ be a local trivialization of a principal G-bundle P. If $U_{\alpha\beta} := U_\alpha \cap U_\beta \neq \varnothing$, then transition functions $g_{\alpha\beta}\colon U_{\alpha\beta} \to \mathrm{Diff}\, G$ are defined by

$$\begin{aligned} \varphi_{\alpha\beta} := \varphi_\alpha \circ \varphi_\beta^{-1}\colon U_{\alpha\beta} \times G &\to U_{\alpha\beta} \times G \\ (x,h) &\mapsto (x, g_{\alpha\beta}(x)h) \end{aligned}$$

Note that

$$\begin{aligned} (\pi(p), g_\alpha(p)) &= \varphi_\alpha \circ \varphi_\beta^{-1} \circ \varphi_\beta(p) \\ &= \varphi_{\alpha\beta}(\pi(p), g_\beta(p)) \end{aligned}$$

This shows

$$g_\alpha(p) = g_{\alpha\beta}(x) g_\beta(p) \tag{2.1}$$

where $p \in \pi^{-1}(x)$. The transition map commutes with the right G-action, so on each fiber it is left multiplication by the uniquely determined element $g_{\alpha\beta}(x) \in G$. Thus the transition functions take values in G:

$$g_{\alpha\beta}\colon U_{\alpha\beta} \to G$$

That is to say, the structure group of a principal G-bundle is G.

2.2.3. *Section.*

Definition 2.2.4 (global section)**.** A global section of a principal G-bundle $\pi\colon P \to M$ is a smooth map $s\colon M \to P$ such that $\pi \circ s = \mathrm{id}$.

Proposition 2.2.3. A principal G-bundle P over M admits a section if and only if it is trivial[2].

Proof. If $s\colon M \to P$ is a smooth section, consider

$$\begin{aligned} \varphi\colon P &\to M \times G \\ p &\mapsto (\pi(p), g(p)) \end{aligned}$$

[2]A vector bundle always has the zero section, but it need not have a nowhere-vanishing section.

where $g(p) \in G$ is the unique element such that $p = s(\pi(p))g(p)$. It always exists since the right action of G is transitive on each fiber and it is unique since the action is free on each fiber. Clearly, it is G-equivariant, since

$$\varphi(ph) = (\pi(ph), g(ph)) = (\pi(p), g(p)h)$$

and the last equality holds since

$$ph = s(\pi(ph))g(ph) = s(\pi(p))g(ph) = pg^{-1}(p)g(ph) \implies h = g^{-1}(p)g(ph)$$

Thus $\varphi\colon P \to M \times G$ is a morphism between principal G-bundles over M, so by Proposition 2.2.2, P is isomorphic to $M \times G$, that is P is a trivial principal G-bundle. □

Example 2.2.3. Although P may not admit a global section, it always admits local sections σ_α over local trivializations $\{U_\alpha, \varphi_\alpha\}$, which is given by

$$\begin{aligned} \sigma_\alpha\colon U_\alpha &\to \pi^{-1}(U_\alpha) \\ x &\mapsto \varphi_\alpha^{-1}(x, e) \end{aligned}$$

Proposition 2.2.4.

$$\sigma_\beta(x) = \sigma_\alpha(x)g_{\alpha\beta}(x)$$

where $x \in U_{\alpha\beta}$.

Proof. A direct computation shows

$$\begin{aligned} \varphi_\beta(\sigma_\alpha(x)g_{\alpha\beta}(x)) &= \varphi_\beta \circ \varphi_\alpha^{-1}(x, e)g_{\alpha\beta}(x) \\ &= (x, g_{\beta\alpha}(x)g_{\alpha\beta}(x)) \\ &= (x, e) \end{aligned}$$

that is $\sigma_\alpha(x)g_{\alpha\beta}(x) = \varphi_\beta^{-1}(x, e) = \sigma_\beta(x)$. □

2.3. **Associated fiber bundle.** Let $\pi\colon P \to M$ be a principal G-bundle and let F be a smooth manifold admitting a smooth left G-action, that is, a group homomorphism $\rho\colon G \to \mathrm{Diff}(F)$.

Proposition 2.1. The set $P \times_\rho F := P \times F/_\sim$, where $(p, f) \sim (p', f')$ if and only if $p' = pg, f' = g^{-1}f$, admits a fiber bundle structure over M with fiber F.

Proof. Consider the map taking an equivalence class $[p, f]$ to $\pi(p)$. To see the local structure, since we already have the local structure of the principal bundle P, i.e. for any $x \in M$, there exists an open set $U_\alpha \ni x$ and $\varphi_\alpha\colon \pi^{-1}(U_\alpha) \to U_\alpha \times G$. Now we define the local trivialization of $P \times_G V$ as

$$\begin{aligned} \varphi_\alpha^F : (P \times_\rho F)|_{U_\alpha} &\to U_\alpha \times F \\ (p, f) &\mapsto (\pi(p), g_\alpha(p)f) \end{aligned}$$

First note that this is well-defined, since

$$(pg, g^{-1}f) \mapsto (\pi(pg), g_\alpha(pg)g^{-1}f) = (\pi(p), g_\alpha(p)gg^{-1}f) = (\pi(p), g_\alpha(p)f)$$

This map gives a diffeomorphism, since g_α is smooth and taking inverse is a smooth operation of Lie groups. □

Remark 2.3.1 (transition functions of associated bundle). The transition functions of the associated bundle can be computed as follows:

$$\begin{aligned}\varphi_\alpha^F \circ (\varphi_\beta^F)^{-1} \colon U_{\alpha\beta} \times F &\to U_{\alpha\beta} \times F\\ (x,f) &\mapsto (x, g_\alpha(p)(g_\beta(p))^{-1} f)\end{aligned}$$

Then by equation (2.1), the transition functions of the associated fiber bundle are exactly $\{\rho(g_{\alpha\beta}^{-1})\}$.

Example 2.3.1 (associated vector bundle)**.** Now let us consider a special case, namely associated vector bundles. Given a representation of G, that is a group homomorphism $\rho\colon G \to \mathrm{GL}(V)$, one can construct a vector bundle $P \times_\rho V$. However, there is a simpler way to construct it from the viewpoint of transition functions: By Remark 2.3.1, we can see the transition functions of this associated vector bundle are $\{\rho(g_{\alpha\beta}^{-1})\}$, where $\{g_{\alpha\beta}\}$ are transition functions of P.

Remark 2.3.2 (relations between vector bundle and principal bundle). For a $\mathrm{GL}(r,\mathbb{R})$-principal bundle P, it provides a vector bundle by considering $P \times_\rho \mathbb{R}^n$, where $\rho\colon \mathrm{GL}(r,\mathbb{R}) \to \mathrm{GL}(r,\mathbb{R})$ is the identity. Conversely, let E be a vector bundle of rank r. Then it provides a $\mathrm{GL}(r,\mathbb{R})$-principal bundle by considering the frame bundle $F(E)$. This shows that there is a one-to-one correspondence

$$\begin{aligned}\mathcal{P}_{\mathrm{GL}(n)}M &\longleftrightarrow \mathrm{Vect}_n^{\mathbb{R}} M\\ P &\mapsto P \times_\rho \mathbb{R}^n.\end{aligned}$$

From this viewpoint, principal G-bundles generalize the concept of vector bundles.

Example 2.3.2. There are two important examples of associated bundles that we will use later.

(1) The associated bundle obtained from conjugation action Conj of G acting on G, denoted by $P \times_{\mathrm{Conj}} G$.
(2) The associated vector bundle obtained from adjoint action $\mathrm{Ad}\colon G \to \mathrm{Aut}(\mathfrak{g})$ is denoted by $\mathrm{ad}(P)$, and it is called the adjoint bundle in the literature.

Example 2.3.3. Let E be a vector bundle of rank r and $F(E)$ be the corresponding $\mathrm{GL}(r,\mathbb{R})$-principal bundle. In this case, the adjoint action is given by

$$\begin{aligned}\mathrm{Ad}\colon \mathrm{GL}(r,\mathbb{R}) &\to \mathrm{GL}(r^2,\mathbb{R})\\ X &\mapsto \{A \mapsto XAX^{-1}\}.\end{aligned}$$

Suppose the transition functions of E are given by $\{g_{\alpha\beta}\}$. Then the transition functions of $\mathrm{ad}(F(E))$ are given by $X \mapsto g_{\alpha\beta}^{-1} X g_{\alpha\beta}$. On the other hand, the transition functions of $\mathrm{End}(E)$ are given by $g_{\alpha\beta}^{-1} \otimes g_{\alpha\beta}$. By the canonical identification

$$g_{\alpha\beta}^{-1} \otimes g_{\alpha\beta}(X) = g_{\alpha\beta}^{-1} X g_{\alpha\beta},$$

we have $\mathrm{ad}(F(E)) \cong \mathrm{End}(E)$.

Proposition 2.3.1. There is a one-to-one correspondence

$$C^\infty(M, P\times_\rho F) \xleftrightarrow{1-1} \{f : P \to F \mid f \text{ is smooth and } f(xg) = g^{-1}f(x)\}$$

Proof. For a G-equivariant smooth function $f: P \to F$, consider $s_f \in C^\infty(M, P\times_\rho F)$ given by

$$s_f(x) = \{(p, f(p)) \mid \pi(p) = x\}$$

where $x \in M$. It is well-defined, since if we choose pg instead of p for some $g \in G$, then $s_f(x) = (pg, f(pg)) = (pg, g^{-1}f(p)) \sim (p, f(p)) \in P\times_\rho F$. Conversely, given $s \in C^\infty(M, P\times_\rho F)$, for any $p \in P$, we consider $\pi(p) = x \in M$ and write $s(x) = [(p, v)]$, then we define $f(p) = v$. It is clear $f(pg) = g^{-1}f(p)$, since $[(p, v)] = [(pg, g^{-1}v)]$. □

Remark 2.3.3. In fact, this proposition is not a coincidence, and it is an important motivation for introducing principal G-bundles. If $\pi: P \to M$ is a principal G-bundle, and E is a vector bundle over M such that E is an associated vector bundle of P, then if we use π to pull E back to P, we claim that the vector bundle π^*E is the trivial bundle $P\times V$ over P. Indeed, we define the following bundle map

$$\begin{aligned} \psi: P\times V &\to P\times_G V \\ (p, v) &\mapsto [p, v] \end{aligned}$$

and consider the following diagram

$$\begin{tikzcd} P\times V \arrow[r] \arrow[d, "\psi"] & P \arrow[d, "\pi"] \\ E = P\times_G V \arrow[r] & M \end{tikzcd}$$

Clearly $P\times V$ satisfies the universal property of pullback, thus by uniqueness we obtain $\pi^*E \cong P\times V$.

In the general case, we can use π to pull $(P\times_G V)\otimes E'$ back to P, and prove that it is $(P\times V)\otimes \pi^*E'$ by the same method. The cases we will encounter are $E' = T^*M$ or $E' = \bigwedge^k T^*M$. We use $\Omega^k_M(P\times_G V)$ to denote $(P\times_G V)\otimes\bigwedge^k T^*M$, the generalization shows that we have the one-to-one correspondence between sections of $\Omega^k_M(P\times_G V)$ and sections of $(P\times V)\otimes\pi^*\bigwedge^k T^*M$ with equivariance conditions. We will call such forms basic forms, a concept we will define in section 4.2.

2.4. **Reduction of principal bundle.** Let $\pi: P \to M$ be a principal G-bundle and let $\pi': P' \to M$. Furthermore, suppose there is a Lie group homomorphism $\alpha: H \to G$.

Definition 2.4.1 (reduction)**.** If there exists a smooth map $\varphi: P' \to P$ such that the following diagram commutes

$$\begin{tikzcd} P' \arrow[rr, "\varphi"] \arrow[dr, "\pi_F"'] & & P \arrow[dl, "\pi_E"] \\ & M & \end{tikzcd}$$

and φ is H-equivariant, that is for any $p' \in P'$ and $h \in H$

$$\varphi(ph) = \varphi(p)\alpha(h)$$

Then P is called an extension of P' from H to G and P' is called a reduction of P from G to H.

Remark 2.4.1. Here are two cases we are concerned about:

(1) $H < G$ is a subgroup, α is an inclusion.
(2) $\alpha : H \to G$ is surjective, for example, H is the universal cover of G.

Extensions of principal bundles always exist, and they are unique, according to the following proposition.

Proposition 2.4.1. Given a Lie group homomorphism $\alpha : H \to G$ and a H-principal bundle P', there exists a unique extension of P' from H to G.

Proof. Existence: Note that $\alpha : H \to G$ gives a smooth left H-action on G. Then consider the associated fiber bundle $P' \times_H G$, it is a principal G-bundle, and if we define

$$\begin{aligned} \varphi : P' &\to P' \times_H G \\ p' &\mapsto [p', 1] \end{aligned}$$

Then φ is desired equivariant map, which makes the diagram commute.

Uniqueness: If there is another extension $\varphi' : P' \to P$, in order to make the following diagram commute

$$\begin{tikzcd} & P' \times_H G \arrow[dd, dashed, "\psi"] \\ P' \arrow[ur, "\varphi"] \arrow[dr, "\varphi'"'] & \\ & P \end{tikzcd}$$

we define ψ by $\psi([p,1]) = \varphi'(p)$. Thus principal G-bundles $P' \times_H G$ and P are isomorphic to each other. $\square$

However, reduction may not exist.

Lemma 2.4.1. Let $\alpha : H \to G$ be a Lie group homomorphism, P be a principal G-bundle with transition functions $\psi_{\alpha\beta} : U_{\alpha\beta} \to G$. The following statements are equivalent:

(1) There exists a reduction of P from G to H.
(2) There exists $\varphi_{\alpha\beta} : U_{\alpha\beta} \to H$ such that $\alpha \circ \varphi_{\alpha\beta} = \psi_{\alpha\beta}$.

Corollary 2.4.1. Let P be a principal G-bundle and H is a Lie subgroup of G, then there exists a reduction of P from G to H if and only if there exist transition functions of P taking values in H.

Example 2.4.1. If $E \to M$ is a complex vector bundle with a Hermitian inner product, then a local trivialization

$$\varphi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{C}^n$$

gives a Hermitian inner product on $\mathbb{C}^n$. Thus a transition function must preserve the inner product, thus

$$\begin{tikzcd} U_\alpha \cap U_\beta \arrow[r] \arrow[dr, dashed] & \mathrm{GL}_n(\mathbb{C}) \\ & \mathrm{U}(n) \arrow[u, hook] \end{tikzcd}$$

This gives a reduction of $\mathrm{GL}_n(\mathbb{C})$-principal bundle to a $\mathrm{U}(n)$-principal bundle.

Example 2.4.2. If $E \to M$ is a real vector bundle, by the same argument we can always reduce its frame bundle P, that is from a $\mathrm{GL}_n(\mathbb{R})$-principal bundle, to a $\mathrm{O}(n)$-principal bundle. Furthermore,

(1) P can be reduced to a $\mathrm{SO}(n)$-principal bundle if and only if E is orientable.
(2) P can be reduced to a $\{e\}$-principal bundle if and only if E is trivial.

Example 2.4.3. Let (M,g) be an oriented Riemannian manifold. Then TM is a $\mathrm{SO}(n)$-principal bundle. Consider the universal cover $\mathrm{Spin}(n) \xrightarrow{2:1} \mathrm{SO}(n)$. If there exists a reduction from $\mathrm{SO}(n)$ to $\mathrm{Spin}(n)$, then we say M admits a spin structure.

3. Connections on principal bundles

3.1. **Forms with values in a vector space.** In this section, let M be a smooth manifold, let V be a vector space with basis $\{e_\alpha\}$ and G a Lie group with Lie algebra $\mathfrak{g}$. A k-form with values in a vector space V can be written as

$$\omega = \omega^\alpha e_\alpha$$

where ω^α are k-forms.

Notation 3.1.1. $\Omega_M^k(V)$ denotes the bundle of k-forms with values in V.

$\Omega_M^k(V)$ is an easy generalization of differential forms, just by replacing $\mathbb{R}$ with a general vector space, and properties of k-forms also hold for k-forms with values in V.

Definition 3.1.1 (exterior derivative)**.** Let $\omega = \omega^\alpha e_\alpha$ be a k-form with values in V, then its exterior derivative is defined as

$$\mathrm{d}\omega = \mathrm{d}\omega^\alpha e_\alpha$$

Proposition 3.1.1 (Cartan's formula)**.** Let $\omega = \omega^\alpha e_\alpha$ be a k-form with values in V, then

$$\begin{aligned}(\mathrm{d}\omega)(X_1,\dots,X_{k+1}) = &\sum_{i=1}^{k+1}(-1)^{i+1}X_i\omega(X_1,\dots,\widehat{X_i},\dots,X_{k+1}) \\ &+\sum_{i<j}(-1)^{i+j}\omega([X_i,X_j],X_1,\dots,\widehat{X_i},\dots,\widehat{X_j},\dots,X_{k+1})\end{aligned}$$

where X_i are vector fields.

Definition 3.1.2 (wedge product)**.** Let ω_1,ω_2 be forms with values in V of degrees k and l, respectively, then

$$(\omega_1\wedge\omega_2)(X_1,\dots,X_{k+l}) := \frac{1}{k!\times l!}\sum_{\sigma\in S_{k+l}}(-1)^{|\sigma|}\omega_1(X_{\sigma(1)},\dots,X_{\sigma(k)})\otimes\omega_2(X_{\sigma(k+1)},\dots,X_{\sigma(k+l)})$$

where X_i are vector fields.

Proposition 3.1.2. Let ω_i, where $i = 1,2,3$, be forms with values in V. Then

(1) $(\omega_1\wedge\omega_2)\wedge\omega_3 = \omega_1\wedge(\omega_2\wedge\omega_3)$.
(2) $\mathrm{d}(\omega_1\wedge\omega_2) = \mathrm{d}\omega_1\wedge\omega_2 + (-1)^{\deg\omega_1}\omega_1\wedge\mathrm{d}\omega_2$.

Definition 3.1.3. Let $T\colon V\to W$ be a linear map between vector spaces, and let ω be a k-form with values in V, then $T\omega$ is a k-form with values in W, which is defined as

$$T\omega(X_1,\dots,X_k) := T(\omega(X_1,\dots,X_k))$$

where X_i are vector fields.

Example 3.1.1. Let ω_1,ω_2 be forms of degrees k and l, respectively, then by our definition one has $\omega_1\wedge\omega_2\in\Omega_M^{k+l}(\mathbb{R}\otimes\mathbb{R})$. It is a little bit different from standard definition of wedge product, since $\omega_1 \wedge \omega_2$ should be a $(k+l)$-form. If we consider

$$\begin{aligned} T\colon \mathbb{R}\otimes\mathbb{R} &\to \mathbb{R} \\ a\otimes b &\mapsto ab \end{aligned}$$

Then $T(\omega_1\wedge\omega_2)$ is a $(k+l)$-form, which coincides with the standard definition, so we just denote $T(\omega_1\wedge\omega_2)$ by $\omega_1\wedge\omega_2$ for convenience.

Example 3.1.2. Let ω_1 be a k-form with values in $\mathfrak{g}$, and ω_2 an l-form with values in V. Given a representation $\rho\colon G\to \mathrm{GL}(V)$, it induces a representation of the Lie algebra, that is $\rho_*\colon \mathfrak{g}\to\mathfrak{gl}(V)$. If we consider

$$\begin{aligned} T\colon \mathfrak{g}\otimes V &\to V \\ \xi\otimes v &\mapsto \rho_*(\xi)v \end{aligned}$$

Then $T(\omega_1\wedge\omega_2)$ is a $(k+l)$-form with values in V, and we denote it by $\omega_1\wedge\omega_2$ for convenience.

Example 3.1.3. Let ω_1,ω_2 be forms with values in $\mathfrak{g}$ of degrees k and l, respectively, by our definition $\omega_1\wedge\omega_2$ is a $(k+l)$-form with values in $\mathfrak{g}$. If we consider

$$\begin{aligned} T\colon \mathfrak{g}\otimes\mathfrak{g} &\to \mathfrak{g} \\ \xi\otimes\eta &\mapsto [\xi,\eta] \end{aligned}$$

Then $T(\omega_1\wedge\omega_2)$ is a $(k+l)$-form with values in $\mathfrak{g}$, and we denote it by $\omega_1\wedge\omega_2$ for convenience.

Remark 3.1.1. If Lie group $G=\mathrm{GL}(n,\mathbb{R})$, then $\mathfrak{g}=\mathfrak{gl}(n,\mathbb{R})$ consists of matrices. Thus in this case for any $\xi,\eta\in\mathfrak{g}$, we can define T as multiplying them together to obtain an element in $\mathfrak{gl}(n,\mathbb{R})$. However, these two notations may cause some misunderstandings.

Example 3.1.4. Let ω be a 1-form with values in $\mathfrak{g}$, then for vector fields X,Y, one has

$$\begin{aligned} \omega\wedge\omega(X,Y) &= T((\omega\wedge\omega)(X,Y)) \\ &= T(\frac{1}{1!\times 1!}(\omega(X)\otimes\omega(Y)-\omega(Y)\otimes\omega(X))) \\ &= [\omega(X),\omega(Y)]-[\omega(Y),\omega(X)] \\ &= 2[\omega(X),\omega(Y)] \end{aligned}$$

Remark 3.1.2. If T is chosen as in Remark 3.1.1, then in this case we have

$$\omega\wedge\omega(X,Y)=[\omega(X),\omega(Y)]$$

So be careful about which wedge product you're using.

Proposition 3.1.3. Let ω be a 1-form with values in $\mathfrak{g}$, then

$$(\omega\wedge\omega)\wedge\omega=\omega\wedge(\omega\wedge\omega)=0$$

Proof. For arbitrary vector fields X,Y and Z, one has

$$\begin{aligned}(\omega\wedge\omega)\wedge\omega(X,Y,Z) &= \frac{1}{2!\times 1!}\{[\omega\wedge\omega(X,Y),\omega(Z)]+[\omega\wedge\omega(Y,Z),\omega(X)]+[\omega\wedge\omega(Z,X),\omega(Y)]\\ &\quad -[\omega\wedge\omega(Y,X),\omega(Z)]+[\omega\wedge\omega(Z,Y),\omega(X)]+[\omega\wedge\omega(X,Z),\omega(Y)]\}\\ &= \frac{2}{2!\times 1!}\{[[\omega(X),\omega(Y)],\omega(Z)]+[[\omega(Y),\omega(Z)],\omega(X)]+[[\omega(Z),\omega(X)],\omega(Y)]\\ &\quad -[[\omega(Y),\omega(X)],\omega(Z)]+[[\omega(Z),\omega(Y)],\omega(X)]+[[\omega(X),\omega(Z)],\omega(Y)]\}\end{aligned}$$

This equals zero according to Jacobi identity of Lie bracket. □

Proposition 3.1.4. Let ω_1,ω_2 be forms with values in $\mathfrak{g}$ of degrees k and l, respectively, then

$$\omega_1\wedge\omega_2=(-1)^{kl+1}\omega_2\wedge\omega_1$$

Proof. Note that for a k-form ω_1 and a l-form ω_2, we have

$$\omega_1\wedge\omega_2=(-1)^{kl}\omega_2\wedge\omega_1$$

But in this case, there is one more -1 coming from Lie bracket. □

3.2. **Maurer-Cartan form.**

Definition 3.2.1 (Maurer-Cartan form)**.** The Maurer-Cartan form θ is a $\mathfrak{g}$-valued 1-form on G, defined by

$$\theta_g:=(L_{g^{-1}})_*$$

where $g\in G$.

Remark 3.2.1. For an arbitrary vector $v\in T_gG$ given by $\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} ge^{tX}$, where $X\in\mathfrak{g}$. A direct computation shows

$$\begin{aligned}\theta_g(v)&=(L_{g^{-1}})_*v\\ &=\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} g^{-1}ge^{tX}\\ &=X\in\mathfrak{g}\end{aligned}$$

This shows that the Maurer-Cartan form is a $\mathfrak{g}$-valued 1-form.

Proposition 3.2.1. Let $G\subseteq \mathrm{GL}(n,\mathbb{R})$ be a matrix Lie group, and let $g: M\to G$ be a smooth map, where M is a smooth manifold. Then $g^*\theta=g^{-1}\mathrm{d}g$, where θ is the Maurer-Cartan form on G and $g^{-1}\mathrm{d}g$ is matrix multiplication.

Proof. For $v\in T_xM$, direct computation shows

$$\begin{aligned}(g^*\theta)_xv&=\theta_{g(x)}((\mathrm{d}g)_xv)\\ &=(L_{g(x)^{-1}})_*(\mathrm{d}g)_xv\end{aligned}$$

Note that

$$\begin{aligned}L_{g(x)^{-1}}:\ \mathrm{GL}(n,\mathbb{R})&\to \mathrm{GL}(n,\mathbb{R})\\ A&\mapsto g(x)^{-1}A\end{aligned}$$

is a linear transformation, which implies $(L_{g(x)^{-1}})_* = L_{g(x)^{-1}}$. Thus

$$(g^*\theta)_x v = g(x)^{-1}(\mathrm{d}g)_x v$$

which implies $g^*\theta = g^{-1}\mathrm{d}g$. □

Corollary 3.2.1. Let $G \subseteq \mathrm{GL}(n,\mathbb{R})$ be a matrix Lie group. Then the Maurer-Cartan form on G is given by $g^{-1}\mathrm{d}g$, where $g\colon G \to G$ is identity map and $g^{-1}\mathrm{d}g$ is matrix multiplication.

3.3. **Motivation for connection on principal bundle.** All in all, our motivation is that connections on principal G-bundles can be used as a tool to study connections on vector bundles E, if E is an associated vector bundle of P.

Recall that a connection on a vector bundle E is defined as the following $\mathbb{R}$-linear operator

$$\nabla\colon C^\infty(M,E) \to C^\infty(M,\Omega^1_M(E))$$

satisfying Leibniz rule.

Suppose E is associated to a principal G-bundle $\pi\colon P \to M$, written as $P \times_\rho V$, then from Proposition 2.3.1, there is a one-to-one correspondence between sections of E and G-equivariant maps from P to V. Given a section s of E, if we use s^P to denote the G-equivariant map obtained from one-to-one correspondence, it is easy to take derivatives of s^P to obtain a 1-form on P with values in V, that is a G-equivariant fiber-wise linear map from TP to V.

However, this 1-form does not by itself define a covariant derivative of s. Since by definition of connection, $\nabla s \in C^\infty(M,\Omega^1_M(E))$, then by Remark 2.3.3, a covariant derivative, when lifted to P, is supposed to be a G-equivariant section over $(P \times V) \otimes \pi^* T^* M$, that is a G-equivariant fiber-wise linear map from $\pi^* TM$ to V.

To see what is missing, it is important to keep in mind that TP has some properties that arise from the fact that P be a principal bundle over M. In fact, we have the following exact sequence

$$0 \to \ker \pi_* \to TP \to \pi^* TM \to 0 \tag{3.1}$$

This exact sequence is quite important. Let us make the following remarks:

Remark 3.3.1. The map from $\ker \pi_*$ to TP is clearly an inclusion. And the surjective map from TP to $\pi^* TM$ is characterized as follows

$$\begin{aligned} TP &\to \pi^* TM \subseteq P \times TM \\ v &\mapsto (p, \pi_* v) \end{aligned}$$

where $v \in T_p P$.

Remark 3.3.2. $\ker \pi_*$ is isomorphic to trivial bundle $P \times \mathfrak{g}$. Indeed, we have the following bundle isomorphism

$$\begin{aligned} \psi\colon P \times \mathfrak{g} &\to \ker \pi_* \\ (p, X) &\mapsto \sigma(X) \end{aligned}$$

where $\sigma(X)_p := \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} pe^{tX}$ is called the fundamental vector field of X. It is clear $\sigma(X) \in \ker \pi_*$, since for each $p \in P$,

$$\begin{aligned}\pi_*(\sigma(X)_p) &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \pi(pe^{tX}) \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \pi(p) \\ &= 0\end{aligned}$$

Remark 3.3.3 (G-equivariance of exact sequence). The action of G on P can be lifted to the exact sequence (3.1). Let $R_g \colon P \to P$ denote the action of $g \in G$ on P, given by $p \mapsto pg$.

(1) The G-action on TP is given by $(R_g)_* \colon TP \to TP$, and it descends to $\ker \pi_*$ since if $v \in \ker \pi_*$, then

$$\begin{aligned}\pi_*((R_g)_* v) &= (\pi \circ R_g)_*(v) \\ &= \pi_*(v) \\ &= 0\end{aligned}$$

(2) The G-action on $\pi^* TM$ is defined by sending a pair $(p, v) \in P \times TM$ to the pair (pg, v). It is well-defined, that is $(pg, v) \in \pi^* TM$, since $\pi(pg) = \pi(p) = \pi(v)$.

Furthermore, we claim the exact sequence (3.1) is equivariant with respect to the lifts.

(1) It automatically holds for inclusion map from $\ker \pi_*$ to TP, since the G-action on $\ker \pi_*$ is obtained by descending the one on TP.
(2) It holds for the map from TP to $\pi^* TM$, since for $v \in TP$ we have $(R_g)_* v$ is sent to $(pg, \pi_*(R_g)_* v)$, that is exactly $(pg, \pi_* v)$, since $\pi \circ R_g = \pi$.

If we want to identify $\ker \pi_*$ as $P \times \mathfrak{g}$, we need to choose an appropriate G-action on $\mathfrak{g}$ so that the isomorphism ψ is G-equivariant. It turns out to be the adjoint representation. Indeed, direct computation shows

$$\begin{aligned}(R_g)_* \psi(p, X) &= (R_g)_* \left(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} p \exp(tX) \right) \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} p \exp(tX) g \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} (pg) \big(g^{-1} \exp(tX) g \big) \\ &= \psi(pg, \mathrm{Ad}(g^{-1}) X)\end{aligned}$$

3.4. **Connection on principal bundle.** So if we want to obtain a fiber-wise linear map $\pi^* TM \to V$ from a fiber-wise linear map $TP \to V$, one way is to require that the exact sequence (3.1) splitting. In other words, we require that there exists a G-equivariant $\omega \colon TP \to P \times \mathfrak{g}$, such that $\omega|_{P \times \mathfrak{g}}$ is identity. Such an ω is called a connection on the principal G-bundle P.

Definition 3.4.1 (connection on a principal G-bundle)**.** Let $\pi\colon P\to M$ be a principal G-bundle. $\omega\in C^\infty(P,\Omega^1_P(\mathfrak{g}))$ is called a connection on P, if it satisfies

(1) For any $X\in\mathfrak{g}$, $\omega(\sigma(X))=X$.
(2) For any $g\in G$, $(R_g)^*\omega=\operatorname{Ad}(g^{-1})\omega$, that is

$$\omega((R_g)_*X)=\operatorname{Ad}(g^{-1})\omega(X)$$

holds for all $X\in C^\infty(P,TP)$.

Notation 3.4.1. $\mathcal{A}(P)$ denotes the set of all connections on P.

Remark 3.4.1 (horizontal distribution viewpoint). If we define $H=\ker\omega$, then

$$TP=H\oplus(P\times\mathfrak{g})$$

such that $(R_g)_*H_p=H_{pg}$. H is called a horizontal distribution and $P\times\mathfrak{g}$ is called a vertical distribution. Conversely, given a horizontal distribution, one can also construct a connection.

Example 3.4.1 (connection on the trivial principal G-bundle)**.** Consider the trivial principal G-bundle $P=M\times G$. Recall we have a Maurer-Cartan form θ, which is a 1-form with values in $\mathfrak{g}$. Then we can use $\pi_G\colon M\times G\to G$ to pull it back to P to obtain a 1-form on P with values in $\mathfrak{g}$, which is called the Maurer-Cartan form on the trivial principal G-bundle, and it is denoted ω_{mc}. Now let us check that ω_{mc} gives a connection on the trivial principal bundle.

(1) For any $X\in\mathfrak{g}$, we have

$$\begin{aligned}\omega_{mc}(\sigma(X))&=\pi_G^*\theta(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}(x,g)e^{tX})\\&=\theta(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}ge^{tX})\\&=(L_{g^{-1}})_*(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}ge^{tX})\\&=\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}e^{tX}\\&=X\end{aligned}$$

(2) It suffices to check $(R_g)^*\theta=\operatorname{Ad}(g^{-1})\theta$ holds for $g\in G$. At a point $h\in G$, for $v\in T_hG$ given by $\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}he^{tX}$, where $X\in\mathfrak{g}$. A direct computation shows

$$\begin{aligned}(R_g)^*\theta_h(v)&=\theta_{hg}(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}he^{tX}g)\\&=\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}(hg)^{-1}he^{tX}g\\&=\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}g^{-1}e^{tX}g\\&=\operatorname{Ad}(g^{-1})\theta_h(v)\end{aligned}$$

Remark 3.4.2. It is clear that $\ker \omega_{mc} = \pi^* TM$, since ω_{mc} is pullback from a 1-form on G, thus in this case

$$TP \cong TM \oplus TG$$

this is exactly the canonical splitting of TP.

3.5. **Gauge group.**

Definition 3.5.1 (gauge transformation)**.** For a principal G-bundle $\pi\colon P \to M$, a gauge transformation is a G-equivariant diffeomorphism $\Phi\colon P \to P$ such that $\pi = \pi \circ \Phi$.

Notation 3.5.1. $\mathcal{G}(P)$ denotes the set of all gauge transformations of P, which forms a group, called the gauge group.

Remark 3.5.1 (terminology). Here we make some clarifications about terminology. A local gauge is a physicist's terminology for the choice of local trivialization, and changes of local trivialization, that is, transition functions, are called gauge transformations. For physicists, the gauge group is exactly the structure group, and the gauge group we define here is sometimes called global gauge group.

Remark 3.5.2 (local expression of gauge transformation). For a gauge transformation Φ, its action on local trivialization $\varphi_\alpha\colon \pi^{-1}(U_\alpha) \to U_\alpha \times G$, given by $\varphi_\alpha(\Phi(p)) = (\pi(p), g_\alpha(\Phi(p)))$, induces a map $\widetilde{\phi}_\alpha\colon \pi^{-1}(U_\alpha) \to G$ by

$$\widetilde{\phi}_\alpha(p) = g_\alpha(\Phi(p)) g_\alpha(p)^{-1}$$

By the G-equivariance of g_α and Φ one has $\widetilde{\phi}$ is G-invariant, which implies $\widetilde{\phi}_\alpha$ descends to U_α, that is one can define $\phi_\alpha\colon U_\alpha \to G$ via $\widetilde{\phi}_\alpha(p) = \phi_\alpha(\pi(p))$. If we consider on the overlaps $x \in U_{\alpha\beta}$ with $p = \pi^{-1}(x)$, then

$$\begin{aligned}
\phi_\alpha(x) &= g_\alpha(\Phi(p)) g_\alpha(p)^{-1} \\
&= g_\alpha(\Phi(p)) g_\beta(\Phi(p))^{-1} g_\beta(\Phi(p)) g_\beta(p)^{-1} g_\beta(p) g_\alpha(p)^{-1} \\
&= g_{\alpha\beta}(x) \phi_\beta(x) g_{\alpha\beta}(x)^{-1}
\end{aligned}$$

This shows $\{\phi_\alpha\}$ defines a global section of associated bundle obtained from the action of G on G by conjugation, that is $P \times_{\mathrm{Conj}} G$ defined in Example 2.3.2. In fact, we have the following one-to-one correspondence.

Proposition 3.5.1. There is a one-to-one correspondence between the group $\mathcal{G}(P)$ and $C^\infty(M, P \times_{\mathrm{Conj}} G)$.

Proof. We have already seen that a gauge transformation can give an element in $C^\infty(M, P \times_{\mathrm{Conj}} G)$. Conversely, by Proposition 2.3.1, there is a one-to-one correspondence between $C^\infty(M, P \times_{\mathrm{Conj}} G)$ and smooth functions $f\colon P \to G$ that are G-equivariant. For such f, consider $\Phi_f\colon P \to P$ given by $\Phi_f(p) = pf(p)$.

(1) $\pi \circ \Phi_f = \pi$, since $\pi \circ \Phi_f(p) = \pi(pf(p)) = \pi(p)$

(2) It is G-equivariant since

$$\begin{aligned}\Phi_f(pg) &= pgf(pg)\\ &= pgg^{-1}f(p)g\\ &= pf(p)g\\ &= \Phi_f(p)g\end{aligned}$$

The two maps we constructed are clearly inverse to each other, giving the desired correspondence. □

Now we show $\mathcal{G}(P)$ acts on $\mathcal{A}(P)$.

Lemma 3.5.1. For any $X \in \mathfrak{g}$ and $\Phi \in \mathcal{G}(P)$, then

$$\Phi_*(\sigma(X)) = \sigma(X)$$

Proof. A direct computation shows

$$\begin{aligned}\Phi_*\sigma(X) &= \Phi_*\Big(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} pe^{tX}\Big)\\ &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \Phi(pe^{tX})\\ &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \Phi(p)e^{tX}\\ &= \sigma(X)\end{aligned}$$

□

Proposition 3.5.2. $\mathcal{G}(P)$ acts on $\mathcal{A}(P)$ via pullback.

Proof. For $\omega \in \mathcal{A}(P)$ and $\Phi \in \mathcal{G}(P)$, we check $\Phi^*\omega \in \mathcal{A}(P)$.

(1) For any $X \in \mathfrak{g}$, we have

$$\begin{aligned}\Phi^*\omega(\sigma(X)) &= \omega(\Phi_*\sigma(X))\\ &= \omega(\sigma(X))\\ &= X\end{aligned}$$

(2) Note that $(R_g)^*\Phi^* = (R_g \circ \Phi)^* = (\Phi \circ R_g)^*$, since Φ is G-equivariant, thus

$$\begin{aligned}(R_g)^*(\Phi^*\omega) &= \Phi^*((R_g)^*\omega)\\ &= \Phi^*(\mathrm{Ad}(g^{-1})\omega)\\ &= \mathrm{Ad}(g^{-1})\Phi^*\omega\end{aligned}$$

□

Remark 3.5.3. Gauge theory is concerned with the orbit space of $\mathcal{G}(P)$, that is, $\mathcal{A}(P)/\mathcal{G}(P)$.

3.6. **Local expression of connection.** Instead of considering a connection 1-form on P, we want to convert it into one on the base manifold M, since we want to use it to study connections on vector bundles over M. To do this, we divide the process into three steps:

(1) Given a connection on the trivial principal G-bundle, identify it with a 1-form on M with values in $\mathfrak{g}$.
(2) Figure out how this correspondence transforms under gauge transformation.
(3) Since a principal G-bundle admits local trivializations, and transition functions can be regarded as gauge transformations, then we reduce the case to the first two steps.

3.6.1. *Baby case.* Fix a trivial principal G-bundle $P = M \times G$ and use the following notation:

(1) $\pi : P \to M$ is the natural projection, given by $p = (x, g) \mapsto x \in M$.
(2) $i : M \to P$ is the natural inclusion, given by $x \mapsto (x, e) \in P$.

Lemma 3.6.1. For any $A \in C^\infty(M, \Omega^1_M(\mathfrak{g}))$, there exists a unique $\widetilde{A} \in C^\infty(P, \Omega^1_P(\mathfrak{g}))$ such that

(1) $i^* \widetilde{A} = A$.
(2) $\widetilde{A}(\sigma(X)) = 0$, where $X \in \mathfrak{g}$.
(3) $(R_g)^* \widetilde{A} = \mathrm{Ad}(g^{-1}) \widetilde{A}$.

Proof. It suffices to construct $\widetilde{A}$ pointwise.

(a) For $p = (x, e) \in M \times G$, we have

$$T_p P = T_x M \oplus \mathfrak{g}$$

Then $\widetilde{A}$ is uniquely determined at this point according to (1) and (2).

(b) At a point $p' = (x, g) \in M \times G$, it is clear that $p' = pg$ and $(R_g)_* : T_p P \to T_{p'} P$ is an isomorphism, For an arbitrary $v \in T_{p'} P$, we may assume $v = (R_g)_* w$ for some $w \in T_p P$, then

$$\begin{aligned} \widetilde{A}_{p'}(v) &= \widetilde{A}_{pg}((R_g)_* w) \\ &= ((R_g)^* \widetilde{A})_p(w) \\ &= \mathrm{Ad}(g^{-1}) \widetilde{A}(w) \end{aligned}$$

□

Proposition 3.6.1. $i^* : \mathcal{A}(P) \to C^\infty(M, \Omega^1_M(\mathfrak{g}))$ is bijective, that is the following diagram commutes

$$\begin{tikzcd} C^\infty(P, \Omega^1_P(\mathfrak{g})) \arrow[r, "i^*"] & C^\infty(M, \Omega^1_M(\mathfrak{g})) \\ \mathcal{A}(P) \arrow[u, hook] \arrow[ur, "1-1"] & \end{tikzcd}$$

Proof. For any $A \in C^\infty(M, \Omega^1_M(\mathfrak{g}))$, by Lemma 3.6.1 we have $\omega_{mc} + \widetilde{A}$ is also a connection on P, thus we consider

$$\begin{aligned} \tau \colon C^\infty(M, \Omega^1_M(\mathfrak{g})) &\to \mathcal{A}(P) \\ A &\mapsto \omega_{mc} + \widetilde{A} \end{aligned}$$

It is clear that τ is surjective, since for any $\omega \in \mathcal{A}(P)$, we have

$$\tau(i^*(\omega - \omega_{mc})) = \omega_{mc} + \omega - \omega_{mc} = \omega$$

Now it suffices to show $i^*\tau = \mathrm{id}$, which implies that τ is injective and hence bijective. Indeed, for $A \in C^\infty(M, \Omega^1_M(\mathfrak{g}))$,

$$i^*\tau(A) = i^*(\omega_{mc} + \widetilde{A}) = i^*\widetilde{A} = A$$

since $i^*\omega_{mc} = 0$. □

3.6.2. *How to glue.* Any gauge transformation Φ on the trivial principal G-bundle $P = M \times G$ can be written as

$$\Phi(x, g) = (x, \phi(x)g)$$

where $\phi \colon M \to G$ is a smooth map.

Proposition 3.6.2. For $\omega \in \mathcal{A}(P)$

$$i^*\Phi^*\omega = \mathrm{Ad}(\phi^{-1})i^*\omega + \phi^*\theta$$

where θ is the Maurer-Cartan form.

Proof. For any $\omega \in \mathcal{A}(P)$, it can be written as $\omega = \omega_{mc} + \widetilde{A}$ according to Proposition 3.6.1. Then

$$\begin{aligned} i^*\Phi^*\omega &= i^*\Phi^*(\omega_{mc} + \widetilde{A}) \\ &\overset{(1)}{=} i^*\Phi^*\pi_G^*\theta + i^*\Phi^*\widetilde{A} \\ &\overset{(2)}{=} \phi^*\theta + i^*\Phi^*\widetilde{A} \end{aligned}$$

where

(1) follows from the definition of Maurer-Cartan form.
(2) follows from $\pi_G \circ \Phi \circ i(x) = \pi_G \circ \Phi(x, e) = \pi_G(x, \phi(x)) = \phi(x)$ for $x \in M$.

Now it suffices to compute $i^*\Phi^*\widetilde{A}$. For $v \in T_xM$, one has

$$\begin{aligned} (i^*\Phi^*\widetilde{A})_x(v) &= \Phi^*\widetilde{A}_{(x,e)}(v, 0) \\ &= \widetilde{A}_{(x,\phi(x))}(v, 0) \\ &= (R_{\phi(x)})^*\widetilde{A}_{(x,e)}(v, 0) \\ &= \mathrm{Ad}(\phi^{-1}(x))(i^*\widetilde{A})_x(v) \end{aligned}$$

Thus we have

$$\begin{aligned} i^*(\Phi^*\omega) &= \phi^*\theta + \mathrm{Ad}(\phi^{-1})i^*\widetilde{A} \\ &\overset{(3)}{=} \phi^*\theta + \mathrm{Ad}(\phi^{-1})i^*\omega \end{aligned}$$

where (3) follows from $i^*\omega_{mc} = 0$ and $\omega = \omega_{mc} + \widetilde{A}$. □

3.6.3. *General case.* Let $\pi\colon P \to M$ be a principal G-bundle with local trivializations $\{U_\alpha, \varphi_\alpha\}$, and $i_\alpha\colon U_\alpha \to U_\alpha \times G$ sends x to (x,e). For a connection $\omega \in \mathcal{A}(P)$, we define $\omega_\alpha := (\varphi_\alpha^{-1})^* \omega|_{\pi^{-1}(U_\alpha)}$, which is a $\mathfrak{g}$-valued 1-form on $U_\alpha \times G$, and

$$A_\alpha := i_\alpha^* \omega_\alpha \in C^\infty(U_\alpha, \Omega^1_{U_\alpha}(\mathfrak{g}))$$

Remark 3.6.1. In Example 2.2.3 we introduced the local section σ_α with respect to local trivialization $\{U_\alpha, \varphi_\alpha\}$, it is clear that $A_\alpha = \sigma_\alpha^*(\omega|_{\pi^{-1}(U_\alpha)})$.

Proposition 3.6.3.

$$\mathcal{A}(P) \xleftrightarrow{1-1} \{(A_\alpha) \in \prod_\alpha C^\infty(U_\alpha, \Omega^1_M(\mathfrak{g})) \mid A_\beta = \mathrm{Ad}(g_{\alpha\beta}^{-1})A_\alpha + g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta}\}$$

Proof. Note that

$$\begin{aligned} \Phi\colon U_{\alpha\beta} \times G &\to U_{\alpha\beta} \times G \\ (x,h) &\mapsto (x, g_{\alpha\beta}(x)h) \end{aligned}$$

gives a gauge transformation of trivial principal G-bundle $U_{\alpha\beta} \times G$. Then for $\omega \in \mathcal{A}(P)$, one has

$$\begin{aligned} i_\beta^* \Phi^* \omega_\alpha &\overset{(1)}{=} \mathrm{Ad}(g_{\alpha\beta}^{-1})(i_\alpha^* \omega_\alpha) + g_{\alpha\beta}^* \theta \\ &\overset{(2)}{=} \mathrm{Ad}(g_{\alpha\beta}^{-1}) A_\alpha + g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta} \end{aligned}$$

where $g_{\alpha\beta}\colon U_{\alpha\beta} \to G$ are transition functions, and

(1) follows from Proposition 3.6.2.
(2) follows from Proposition 3.2.1.

Note that

$$\begin{aligned} \omega_\alpha &= (\varphi_\alpha^{-1})^* \omega|_{\pi^{-1}(U_\alpha)} \\ &= (\varphi_\alpha^{-1})(\varphi_\beta)^* (\varphi_\beta^{-1})^* \omega|_{\pi^{-1}(U_\alpha)} \\ &= (\varphi_\beta \circ \varphi_\alpha^{-1})^* \omega_\beta \\ &= (\Phi^{-1})^* \omega_\beta \end{aligned}$$

This shows

$$i_\beta^* \Phi^* \omega_\alpha = i_\beta^* \omega_\beta = A_\beta$$

Conversely, suppose $\{A_\alpha\}$ is a set of $\mathfrak{g}$-valued 1-forms satisfying

$$A_\beta = \mathrm{Ad}(g_{\alpha\beta}^{-1}) A_\alpha + g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta}$$

By Lemma 3.6.1 there exists a $\mathfrak{g}$-valued 1-form $\widetilde{A}_\alpha$ on $\pi^{-1}(U_\alpha)$ such that

(1) $(\sigma_\alpha)^* \widetilde{A}_\alpha = A_\alpha$.
(2) $\widetilde{A}_\alpha(\sigma(X)) = 0$ for $X \in \mathfrak{g}$.
(3) $(R_g)^* \widetilde{A}_\alpha = \mathrm{Ad}(g^{-1})\widetilde{A}_\alpha$.

For each α, let $\omega_\alpha = \omega_{mc} + \widetilde{A}_\alpha$ on $U_\alpha \times G$. The transformation law for the A_α is exactly the condition that the local forms ω_α agree under the transition maps. Hence they glue to a global connection ω on P. These two constructions are inverse to each other. $\square$

Corollary 3.6.1. $\mathcal{A}(P)$ is an affine space modelled on $C^\infty(M, \Omega^1_M(\mathrm{ad}(P)))$.

4. Curvature of principal bundle

4.1. Definition.

Definition 4.1.1 (curvature)**.** Let P be a principal G-bundle and $\omega \in \mathcal{A}(P)$. The curvature of ω is defined as

$$\Omega := \mathrm{d}\omega + \frac{1}{2}\omega \wedge \omega \in C^\infty(P, \Omega_P^2(\mathfrak{g}))$$

Proposition 4.1.1.

$$(R_g)^*\Omega = \mathrm{Ad}(g^{-1})\Omega$$

where $g \in G$.

Proof. It follows from the fact that pullback commutes with exterior derivative and wedge product. □

Proposition 4.1.2. Let $P = M \times G$ be the trivial principal G-bundle equipped with a connection ω_{mc}, then $\Omega = 0$.

Proof. It suffices to check that the Maurer-Cartan form $\theta \in C^\infty(G, \Omega_G^1(\mathfrak{g}))$ satisfying

$$\mathrm{d}\theta + \frac{1}{2}\theta \wedge \theta = 0$$

which is called the Maurer-Cartan equation. Firstly we suppose X, Y are left-invariant vector fields, then

$$\theta(X) = (L_{g^{-1}})_* X_g = (L_{g^{-1}})_*(L_g)_* X_e = X_e$$

is constant. Thus

$$\mathrm{d}\theta(X,Y) = -\theta([X,Y]) = -\frac{1}{2}\theta \wedge \theta(X,Y)$$

since $X(\theta(Y)) = Y(\theta(X)) = 0$. But left-invariant vector fields span the tangent space at every point; therefore, the Maurer-Cartan equation holds for arbitrary vector fields X, Y. □

Theorem 4.1.1 (Bianchi identity)**.**

$$\mathrm{d}\Omega + \omega \wedge \Omega = 0$$

Proof.

$$\begin{aligned}
\mathrm{d}\Omega &= \mathrm{d}(\mathrm{d}\omega + \frac{1}{2}\omega \wedge \omega) \\
&= \frac{1}{2}\mathrm{d}\omega \wedge \omega - \frac{1}{2}\omega \wedge \mathrm{d}\omega \\
&= -\omega \wedge \mathrm{d}\omega \\
&= -\omega \wedge (\Omega - \frac{1}{2}\omega \wedge \omega) \\
&= -\omega \wedge \Omega
\end{aligned}$$

□

Definition 4.1.2 (horizontal form)**.** Let α be a k-form on P with values in a vector space V. It is called horizontal if $\iota_{\sigma(X)}\alpha = 0$ for arbitrary $X \in \mathfrak{g}$.

Lemma 4.1.1. For $X \in \mathfrak{g}$, the flow of $\sigma(X)$ is given by

$$\phi_t(p) = pe^{tX}$$

where $p \in P$.

Proposition 4.1.3. Ω is a horizontal 2-form.

Proof. A direct computation shows

(1) For $X, Y \in \mathfrak{g}$, one has

$$\begin{aligned} \mathrm{d}\omega(\sigma(X),\sigma(Y)) &= \sigma(X)(\omega(\sigma(Y))) - \sigma(Y)(\omega(\sigma(X))) - \omega([\sigma(X),\sigma(Y)]) \\ &\overset{(1)}{=} -[\omega(\sigma(X)),\omega(\sigma(Y))] \\ &\overset{(2)}{=} -\frac{1}{2}\omega\wedge\omega(\sigma(X),\sigma(Y)) \end{aligned}$$

where
(1) follows from $\omega(\sigma(Y))$ and $\omega(\sigma(X))$ are constant functions with values Y and X, respectively.
(2) follows from Example 3.1.3.

(2) If $X \in \mathfrak{g}$ and Y is a horizontal vector field, note that

$$\frac{1}{2}\omega\wedge\omega(\sigma(X),Y) = 0$$

since $\omega(Y) = 0$, and direct computation shows

$$\begin{aligned} \mathrm{d}\omega(\sigma(X),Y) &= \sigma(X)(\omega(Y)) - Y\omega(\sigma(X)) - \omega([\sigma(X),Y]) \\ &\overset{(3)}{=} -\omega([\sigma(X),Y]) \\ &\overset{(4)}{=} -\omega(\mathcal{L}_{\sigma(X)}Y) \end{aligned}$$

where
(3) follows from $\omega(Y) = 0$ and $\omega(\sigma(X))$ is a constant function with value X.
(4) follows from the property of Lie derivative.
By definition one has

$$(\mathcal{L}_{\sigma(X)}Y)_p = \lim_{t\to 0}\frac{(\phi_{-t})_* Y_{\phi_t(p)} - Y_p}{t}$$

where ϕ_t is the flow generated by $\sigma(X)$ and $p \in P$. Thus

$$\begin{aligned} \omega_p((\mathcal{L}_{\sigma(X)}Y)_p) &\overset{(5)}{=} \omega_p(\lim_{t\to 0}\frac{(\phi_{-t})_* Y_{\phi_t(p)} - Y_p}{t}) \\ &\overset{(6)}{=} \lim_{t\to 0}\frac{1}{t}\left\{\omega_p((\phi_{-t})_* Y_{\phi_t(p)}) - \omega_p(Y_p)\right\} \\ &\overset{(7)}{=} \lim_{t\to 0}\frac{1}{t}\left\{((R_{e^{-tX}})^*\omega_p)(Y_{pe^{tX}}) - \omega_p(Y_p)\right\} \\ &= \lim_{t\to 0}\frac{1}{t}\left\{\mathrm{Ad}(e^{tX})\omega_{pe^{tX}}(Y_{pe^{tX}}) - \omega_p(Y_p)\right\} \\ &= 0 \end{aligned}$$

where

(5) follows from the definition of Lie derivative.
(6) follows since ω is a smooth form.
(7) follows from Lemma 4.1.1.

$\square$

Remark 4.1.1 (curvature vanishes and integrability). Given a horizontal distribution $H \subseteq TP$, we define the horizontal projection $h\colon TP \to TP$ to be the projection onto the horizontal distribution along the a vertical distribution. Since both vertical and horizontal distributions are invariant under the action of G, so is h. Then $\Omega = h^*\mathrm{d}\omega$. Indeed, it suffices to show for vector fields X, Y, one has

$$\mathrm{d}\omega(hX, hY) = \mathrm{d}\omega(X, Y) + \frac{1}{2}\omega \wedge \omega(X, Y)$$

Consider the following cases:

(1) Let X, Y be horizontal. In this case there is nothing to prove, since $\omega(X) = \omega(Y) = 0$ and $hX = X, hY = Y$.
(2) If one of X, Y is vertical, then it is clear both sides are zero, since both Ω and $h^*\mathrm{d}\omega$ are horizontal.

As a consequence one has

$$\begin{aligned}\Omega(X, Y) &= \mathrm{d}\omega(hX, hY)\\ &= -\omega([hX, hY])\end{aligned}$$

where X, Y are two vector fields on P. This shows $\Omega(X, Y) = 0$ if and only if $[hX, hY]$ is horizontal. In other words, the curvature of the connection measures the failure of integrability of the horizontal distribution $H \subseteq TP$.

4.2. **Local expression of curvature and basic form.** Let $\pi\colon P \to M$ be a principal G-bundle with local trivializations $\{U_\alpha, \varphi_\alpha\}$. If we define

$$\Theta_\alpha = \sigma_\alpha^*(\Omega|_{\pi^{-1}(U_\alpha)}) \in C^\infty(U_\alpha, \Omega^2_{U_\alpha}(\mathfrak{g}))$$

By definition one has

$$\Theta_\alpha = \mathrm{d}A_\alpha + \frac{1}{2}A_\alpha \wedge A_\alpha$$

Lemma 4.2.1. For $x \in U_{\alpha\beta}$ and $v \in T_xM$

$$(\sigma_\beta)_*(v) = (R_{g_{\alpha\beta}(x)})_*((\sigma_\alpha)_* v) + (\sigma_\alpha(x))_*((g_{\alpha\beta})_* v)$$

where $(\sigma(x))_*$ is the differential of the following map

$$\begin{aligned} G &\to P\\ h &\mapsto \sigma_\alpha(x)h\end{aligned}$$

Proof. Let $\gamma(t)$ be a curve with $\gamma(0)=x$ and $\gamma'(0)=v$. A direct computation shows

$$\begin{aligned}(\sigma_\beta)_*(v) &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}\sigma_\beta(\gamma(t))\\ &\overset{(1)}{=} \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}\sigma_\alpha(\gamma(t))g_{\alpha\beta}(\gamma(t))\\ &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}\sigma_\alpha(\gamma(t))g_{\alpha\beta}(x)+\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}\sigma_\alpha(x)g_{\alpha\beta}(\gamma(t))\\ &= (R_{g_{\alpha\beta}(x)})_*((\sigma_\alpha)_*v)+(\sigma_\alpha(x))_*((g_{\alpha\beta})_*v)\end{aligned}$$

where (1) follows from Proposition 2.2.4. □

Remark 4.2.1. From the above proof, it is clear that $(\sigma_\alpha(x))_*((g_{\alpha\beta})_*v)$ is a vertical vector, which is a crucial property.

Proposition 4.2.1.

$$\Theta_\beta = \mathrm{Ad}(g_{\alpha\beta}^{-1})\Theta_\alpha$$

where $g_{\alpha\beta}\colon U_{\alpha\beta}\to G$ is a transition function.

Proof. For $x\in U_{\alpha\beta}$ and $v,w\in T_xM$, direct computation shows

$$\begin{aligned}(\Theta_\beta)_x(v,w) &= \Omega_{\sigma_\beta(x)}((\sigma_\beta)_*v,(\sigma_\beta)_*w)\\ &\overset{(1)}{=} \Omega_{\sigma_\beta(x)}((R_{g_{\alpha\beta}(x)})_*(\sigma_\alpha)_*v,(R_{g_{\alpha\beta}(x)})_*(\sigma_\alpha)_*w)\\ &= ((R_{g_{\alpha\beta}(x)})^*\Omega)_{\sigma_\alpha(x)}((\sigma_\alpha)_*v,(\sigma_\alpha)_*w)\\ &\overset{(2)}{=} \mathrm{Ad}(g_{\alpha\beta}(x)^{-1})\Omega_{\sigma_\alpha(x)}((\sigma_\alpha)_*v,(\sigma_\alpha)_*w)\\ &= \mathrm{Ad}(g_{\alpha\beta}(x)^{-1})(\Theta_\alpha)_x(v,w)\end{aligned}$$

where

(1) follows because Ω is horizontal and remark of Lemma 4.2.1.
(2) follows from Proposition 4.1.1.

□

Definition 4.2.1 (basic form)**.** Let $\rho\colon G\to\mathrm{GL}(V)$ be a representation of G, a k-form α on P with values in V is called a basic form, if it satisfies

(1) α is horizontal.
(2) It is ρ-equivariant, that is

$$(R_g)^*\alpha = \rho(g^{-1})\alpha$$

where $g\in G$.

Notation 4.2.1. The set of all basic k-forms on P with values in V is denoted by $C^\infty(P,\Omega^k_P(V))^{\mathrm{basic}}$.

Theorem 4.2.1. Let $\rho\colon G\to\mathrm{GL}(V)$ be a linear representation, and $E=P\times_\rho V$. Then

$$C^\infty(M,\Omega^k_M(E))\xleftrightarrow{1-1}C^\infty(P,\Omega^k_P(V))^{\mathrm{basic}}$$

Example 4.2.1. For $k = 0$, one has

$$C^\infty(P, \Omega^0_P(V))^{\mathrm{basic}} = \{f : P \to V \mid f(xg) = \rho(g^{-1}) f(x)\}$$

Thus Theorem 4.2.1 recovers Proposition 2.3.1.

5. From connection on principal to connection on vector bundle

5.1. **Connection on vector bundle.** Let $\pi\colon E \to M$ be a vector bundle of rank n, and let $\{U_\alpha, \varphi_\alpha\}$ be a local trivialization of E with transition functions $\{g_{\alpha\beta}\}$. If $\{e_i\}$ is the standard basis of $\mathbb{R}^n$ consisting of column vectors[3], then there is a local frame over U_α given by

$$e_i^\alpha := \varphi_\alpha^{-1}((x, e_i))$$

A direct computation shows

$$\begin{aligned} e_i^\beta &= \varphi_\alpha^{-1} \circ \varphi_\alpha \circ \varphi_\beta^{-1}((x, e_i)) \\ &= \varphi_\alpha^{-1}((x, g_{\alpha\beta} e_i)) \\ &= (g_{\alpha\beta})_i^j e_j^\alpha \end{aligned}$$

where j is the column index and i is the row index of $(g_{\alpha\beta})_i^j$. Let ∇ be a connection on E, which is locally given by $\{A_\alpha\} \in \prod C^\infty(U_\alpha, \Omega_M^1(\mathfrak{gl}(n,\mathbb{R})))$, that is

$$\nabla e_i^\alpha = (A_\alpha)_i^j \otimes e_j^\alpha$$

A direct computation shows

$$\begin{aligned} \nabla e_i^\beta &= \nabla((g_{\alpha\beta})_i^j e_j^\alpha) \\ &= \mathrm{d}(g_{\alpha\beta})_i^j \otimes e_j^\alpha + (g_{\alpha\beta})_i^j (A_\alpha)_j^k \otimes e_k^\alpha \\ &= (\mathrm{d}(g_{\alpha\beta})_i^k + (g_{\alpha\beta})_i^j (A_\alpha)_j^k) \otimes e_k^\alpha \end{aligned}$$

On the other hand, one has

$$\begin{aligned} \nabla e_i^\beta &= (A_\beta)_i^j \otimes e_j^\beta \\ &= (A_\beta)_i^j (g_{\alpha\beta})_j^k \otimes e_k^\alpha \end{aligned}$$

This shows

$$(A_\beta)_i^j (g_{\alpha\beta})_j^k = \mathrm{d}(g_{\alpha\beta})_i^k + (g_{\alpha\beta})_i^j (A_\alpha)_j^k$$

and in matrix notation one has

$$A_\beta = g_{\alpha\beta}^{-1} A_\alpha g_{\alpha\beta} + g_{\alpha\beta}^{-1} \mathrm{d} g_{\alpha\beta} \tag{5.1}$$

That is to say, if we want to give a connection on E, it suffices to give $\{A_\alpha\} \in \prod C^\infty(U_\alpha, \Omega_M^1(\mathfrak{gl}(n,\mathbb{R})))$ satisfying relation (5.1).

5.2. **Connection on associated vector bundle.** In this section we will show that if E is an associated vector bundle of principal G-bundle P over M, then connection ω on P gives a connection on E.

[3] To be explicit, $e_i = (0, \ldots, \underbrace{1}_{i\text{-th}}, \ldots, 0)^T$.

5.2.1. *Baby version.* Let E be a vector bundle over M, and it is realized as an associated vector bundle of principal $\mathrm{GL}(n,\mathbb{R})$-bundle P by trivial representation. Let $\{U_\alpha\}$ be a local trivialization of P with transition functions $\{g_{\alpha\beta}\}$. For connection $\omega \in \mathcal{A}(P)$, by Proposition 3.6.3 one has a set of $\mathfrak{gl}(n,\mathbb{R})$-valued 1-forms $\{A_\alpha\}$ with

$$A_\beta = \mathrm{Ad}(g_{\alpha\beta}^{-1})A_\alpha + g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta}$$

Note that A_α is a 1-form with values in $\mathfrak{gl}(n,\mathbb{R})$, and for matrix groups, the adjoint representation can be expressed explicitly, that is

$$\mathrm{Ad}(g_{\alpha\beta}^{-1})A_\alpha = g_{\alpha\beta}^{-1}A_\alpha g_{\alpha\beta}$$

This shows $\{A_\alpha\}$ which is obtained from ω satisfies relation (5.1), and thus it gives a connection on E.

5.2.2. *General case.* Let P be a principal G-bundle with local trivializations $\{U_\alpha\}$ and transition functions $\{g_{\alpha\beta}\}$, and suppose $E = P\times_\rho \mathbb{R}^n$ is an associated vector bundle given by a representation $\rho\colon G\to \mathrm{GL}(n,\mathbb{R})$. For connection $\omega\in \mathcal{A}(P)$, by Proposition 3.6.3 one has a set of $\mathfrak{g}$-valued 1-forms $\{A_\alpha\}$ with

$$A_\alpha = \mathrm{Ad}(g_{\alpha\beta}^{-1})A_\beta + g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta}$$

Let $\rho_*\colon \mathfrak{g}\to\mathfrak{gl}(n,\mathbb{R})$ be the differential of ρ, and note that the following diagram commutes

$$\begin{tikzcd}
G \arrow[r, "\mathrm{Ad}"] \arrow[d, "\rho"] & \mathrm{Aut}(\mathfrak{g}) \arrow[d, "\rho_*"] \\
\mathrm{GL}(n,\mathbb{R}) \arrow[r, "\mathrm{Ad}"] & \mathfrak{gl}(n,\mathbb{R})
\end{tikzcd}$$

Then $\{\rho_*(A_\alpha)\}$ is a set of $\mathfrak{gl}(n,\mathbb{R})$-valued 1-forms satisfying

$$\begin{aligned}
\rho_*(A_\alpha) &= \rho_*(\mathrm{Ad}(g_{\alpha\beta}^{-1})A_\beta) + \rho_*(g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta})\\
&= \rho(g_{\alpha\beta})^{-1}\rho_*(A_\beta)\rho(g_{\alpha\beta}) + \rho_*(g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta})\\
&= \rho(g_{\alpha\beta})^{-1}\rho_*(A_\beta)\rho(g_{\alpha\beta}) + \rho(g_{\alpha\beta})^{-1}\mathrm{d}\rho(g_{\alpha\beta})
\end{aligned}$$

This shows $\{\rho_*(A_\alpha)\}$ gives a connection on E, since the transition functions[4] of E is $\{\rho(g_{\alpha\beta})\}$.

[4] See Remark 2.3.1.

6. Flat connection and holonomy

6.1. **Lifting of curves.** Let $\pi\colon P\to M$ be a principal G-bundle equipped with a connection ω, consider a smooth curve $\gamma\colon[0,1]\to M$ and a point $p\in\pi^{-1}(\gamma(0))$, we claim there exists a unique smooth map $\widetilde{\gamma}\colon[0,1]\to P$ such that

(1) The following diagram commutes:

$$\begin{tikzcd} & P \arrow[d, "\pi"] \\ {[0,1]} \arrow[r, "\gamma"] \arrow[ur, dashed, "\widetilde{\gamma}"] & M \end{tikzcd}$$

(2) $\widetilde{\gamma}'(t)$ is horizontal.
(3) $\widetilde{\gamma}(0)=p$.

Proof. For convenience we assume G is a matrix group, and without loss of generality, we may assume P is a trivial principal G-bundle $M\times G$, since it is a local problem. In this case we write $\widetilde{\gamma}=(\gamma(t),g(t))$, it is clear that $\pi\circ\widetilde{\gamma}=\gamma$. For conditions (2) and (3), this is in fact an initial-value problem for an ODE: Note that we can write connection $\omega=\omega_{mc}+\widetilde{A}$, so $\widetilde{\gamma}'(t)$ is horizontal if and only if

$$\begin{aligned}(\omega_{mc}+\widetilde{A})(\widetilde{\gamma}'(t))&=(\omega_{mc}+\widetilde{A})((\gamma'(t),g'(t)))\\&=g^{-1}(t)g'(t)+\widetilde{A}((\gamma'(t),g'(t)))\\&=g^{-1}(t)g'(t)+\mathrm{Ad}(g^{-1}(t))A_{\gamma(t)}(\gamma'(t))\\&=g^{-1}(t)g'(t)+g^{-1}(t)A_{\gamma(t)}(\gamma'(t))g(t)\\&=0\end{aligned}$$

This completes the proof. □

6.2. **Flat connection.**

Definition 6.2.1 (flat connection)**.** Let P be a principal G-bundle, a connection $\omega\in\mathcal{A}(P)$ is called flat, if its curvature form $\Omega=0$.

Theorem 6.2.1. The following statements are equivalent:

(1) ω is flat.
(2) There exists a local trivialization $\{U_\alpha,\varphi_\alpha\}$ such that $\omega|_{\pi^{-1}(U_\alpha)}=(\varphi_\alpha)^*\omega_{mc}$.

Hint. The curvature vanishes if and only if the horizontal distribution is integrable. □

Corollary 6.2.1. The following statements are equivalent:

(1) There is a flat connection on P.
(2) There are local trivializations $\{U_\alpha,\varphi_\alpha\}$ such that transition functions $\{g_{\alpha\beta}\colon U_\alpha\cap U_\beta\to G\}$ are locally constant functions.

Proof. From (2) to (1). By Proposition 3.6.3 a connection $\omega\in\mathcal{A}(P)$ is given by $\{A_\alpha\}$ such that

$$A_\beta=\mathrm{Ad}(g_{\alpha\beta}^{-1})A_\alpha+g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta}$$

If $g_{\alpha\beta}$ are locally constant functions, then $\mathrm{d}g_{\alpha\beta}=0$, and thus $A_\alpha=0$ gives a flat connection.

From (1) to (2). If ω is a flat connection, by Theorem 6.2.1 there exists a local trivialization $\{U_\alpha,\varphi_\alpha\}$ of P such that $\omega|_{\pi^{-1}(U_\alpha)}$ is $(\varphi_\alpha)^*\omega_{mc}$. Then with respect to this local trivialization, one has

$$A_\alpha=(\sigma_\alpha)^*(\varphi_\alpha)^*\omega_{mc}=0$$

for all α. This shows $g^{-1}_{\alpha\beta}\mathrm{d}g_{\alpha\beta}=0$ for all α,β, that is $g_{\alpha\beta}$ are locally constant functions. □

Corollary 6.2.2. Flat connections are equivalent to $\mathbb{R}$-valued local systems.

6.3. **Holonomy and Riemann-Hilbert correspondence.** Let $\gamma\colon[0,1]\to M$ be a smooth closed curve with lifting $\widetilde{\gamma}\colon[0,1]\to P$ starting at $\widetilde{\gamma}(0)\in\pi^{-1}(\gamma(0))$. Note that

$$\widetilde{\gamma}(1)\in\pi^{-1}(\gamma(1))=\pi^{-1}(\gamma(0))$$

So there exists $g\in G$ such that $\widetilde{\gamma}(1)=\widetilde{\gamma}(0)g$, since the fiber is a G-orbit of G. The element g is called the holonomy, which is denoted by $\mathrm{Hol}(\gamma,p)$, since it only depends on γ and p.

Proposition 6.3.1.

(1) For $p,pg\in P$, where $g\in G$, one has

$$\mathrm{Hol}(\gamma,pg)=g^{-1}\,\mathrm{Hol}(\gamma,p)g$$

(2) Let γ_1,γ_2 be two smooth closed curves, then

$$\mathrm{Hol}(\gamma_1\gamma_2,p)=\mathrm{Hol}(\gamma_1,p)\mathrm{Hol}(\gamma_2,p)$$

Proof. It is clear. □

From part (2) of the above proposition, Hol can be regarded as a group homomorphism to some extent, so if we want to give a homomorphism

$$\mathrm{Hol}\colon\pi_1(M)\to G$$

It suffices to check when $\mathrm{Hol}(\gamma,p)$ is independent of homotopy class. Consider the following homotopy

$$\gamma_s\colon(-\varepsilon,\varepsilon)\times[0,1]\to M$$

such that $\gamma_0=\gamma$. If we write its lift in a local trivialization as $\widetilde{\gamma}_s(t)=(\gamma_s(t),g_s(t))$, then the following equation holds

$$\frac{\partial g_s}{\partial t}(t)+A_{\gamma(t)}(\frac{\partial g_s}{\partial t}(t))g_s(t)=0$$

So if ω is a flat connection, then it reduces to the statement that, for arbitrary $s\in(-\varepsilon,\varepsilon)$, one has $\frac{\partial g_s}{\partial t}(t)=0$. This shows it is independent of s.

Theorem 6.3.1 (Riemann-Hilbert correspondence)**.**

$$\{\text{flat connections on } P\}/\text{isomorphism}\xleftrightarrow{1-1}\mathrm{Hom}(\pi_1(M),G)/\text{conjugacy}$$

Part 2. Chern-Weil theory

7. Chern-Weil homomorphism

7.1. Invariant polynomial.

7.1.1. *General theory.* Let V be a vector space over $\mathbb{R}$ and $\mathrm{Sym}^k V^*$ the space of symmetric k-linear maps f from $V\times\cdots\times V$ to $\mathbb{R}$, and $\mathrm{Sym}\, V^* = \bigoplus_{k=0}^{\infty} \mathrm{Sym}^k V^*$ is a commutative algebra over $\mathbb{R}$, where the multiplication is given by

$$fg(x_1,\dots,x_{k+l}) = \frac{1}{(k+l)!}\sum_{\sigma\in S_{k+l}} f(x_{\sigma(1)},\dots,x_{\sigma(k)})g(x_{\sigma(k+1)},\dots,x_{\sigma(k+l)})$$

where $f\in \mathrm{Sym}^k V^*, g\in \mathrm{Sym}^l V^*$ and $x_i\in V$. Let $P^k(V)$ denote the space of homogeneous polynomial functions of degree k on V. Then $P(V)=\bigoplus_{k=0}^{\infty} P^k(V)$ is the algebra of polynomial functions on V.

Proposition 7.1.1. The mapping $\varphi\colon \mathrm{Sym}\, V^* \to P(V)$ defined by

$$(\varphi f)(t) := f(t,\dots,t)$$

for $f\in \mathrm{Sym}^k(V)$ and $t\in V$ is an isomorphism.

Proof. See Proposition 2.1 in [KN96]. □

Proposition 7.1.2. Given a group of linear transformations of V, let $\mathrm{Sym}_G V^*$ and $P_G(V)$ be the subalgebra of $\mathrm{Sym}\, V^*$ and $P(V)$, respectively, consisting of G-invariant elements. Then the isomorphism in Proposition 7.1.1 gives an isomorphism from $\mathrm{Sym}_G V^*$ to $P_G(V)$.

Proof. The proof is straightforward and is left to the reader. □

7.1.2. *G-invariant polynomial.* Let G be a Lie group with Lie algebra $\mathfrak{g}$ and let $\mathrm{Sym}^k \mathfrak{g}^*$ be the symmetric k-linear functionals, that is,

$$\mathrm{Sym}^k \mathfrak{g}^* = \{f : \underbrace{\mathfrak{g}\times\cdots\times\mathfrak{g}}_{k \text{ times}} \to \mathbb{R} \mid f \text{ is } k\text{-linear and symmetric}\}$$

Furthermore, G acts on $\mathrm{Sym}^k \mathfrak{g}^*$ as follows

$$gf(x_1,\dots,x_k) := f(\mathrm{Ad}(g)x_1,\dots,\mathrm{Ad}(g)x_k)$$

where $f\in \mathrm{Sym}^k \mathfrak{g}^*$ and $x_1,\dots,x_k\in\mathfrak{g}$.

Definition 7.1.1 (G-invariant polynomial)**.** The set of G-invariant polynomials of degree k is

$$I^k(\mathfrak{g}) := \{f\in \mathrm{Sym}^k \mathfrak{g}^* \mid gf = f, \forall g\in G\}$$

and

$$I(\mathfrak{g}) := \bigoplus_{k\geq 0} I^k(\mathfrak{g})$$

which is a commutative algebra over $\mathbb{R}$.

Proposition 7.1.3. The algebra $I(\mathfrak{g})$ may be identified with the algebra of $\mathrm{Ad}(G)$-invariant polynomial functions on $\mathfrak{g}$.

7.2. **Chern-Weil homomorphism.** Let $\pi\colon P \to M$ be a principal G-bundle, and let ω be a connection on P with curvature Ω.

Lemma 7.2.1. Let $\widetilde{\alpha}$ be a k-form on P such that

(1) $R_g^*(\widetilde{\alpha}) = \widetilde{\alpha}$.
(2) $\widetilde{\alpha}$ is horizontal.

Then there exists a unique k-form α on M such that $\widetilde{\alpha} = \pi^*\alpha$.

Proof. For any vector fields $X_1,\dots,X_k$ on M, there are R_g-invariant vector fields $\widetilde{X}_1,\dots,\widetilde{X}_k$ such that $\pi_*(\widetilde{X}_i) = X_i$, where $i = 1,\dots,k$. Given $\widetilde{\alpha}$ as above, we define

$$\alpha(X_1,\dots,X_k) = \widetilde{\alpha}(\widetilde{X}_1,\dots,\widetilde{X}_k)$$

It suffices to check it is well-defined, that is it is independent of the choice of $\widetilde{X}_1,\dots,\widetilde{X}_k$. Indeed, suppose $\widetilde{X}'_1,\dots,\widetilde{X}'_k$ are other R_g-invariant vector fields with $\pi_*(\widetilde{X}'_i) = X_i$, where $i = 1,\dots,k$. Then

$$\begin{aligned}\widetilde{\alpha}(\widetilde{X}_1,\dots,\widetilde{X}_k) - \widetilde{\alpha}(\widetilde{X}'_1,\dots,\widetilde{X}'_k) &= \widetilde{\alpha}(\widetilde{X}_1 - \widetilde{X}'_1,\dots,\widetilde{X}_k - \widetilde{X}'_k)\\ &= 0\end{aligned}$$

since $\widetilde{X}_i - \widetilde{X}'_i$ is vertical for $i = 1,\dots,k$. The uniqueness follows from the fact that π is a submersion. $\square$

Proposition 7.2.1. For $f \in I^k(\mathfrak{g})$, one has

$$f(\Omega) := f(\underbrace{\Omega \wedge \cdots \wedge \Omega}_{k \text{ times}})$$

is a $2k$-form[5] on P, and

(1) $f(\Omega)$ is horizontal, G-invariant and closed.
(2) There exists a unique $2k$-form $f(\Theta)$ on M such that $\pi^*(f(\Theta)) = f(\Omega)$ and $\mathrm{d}f(\Theta) = 0$.
(3) $[f(\Theta)] \in H^{2k}(M,\mathbb{R})$ is independent of the choice of connection ω.

Proof. For (1). $f(\Omega)$ is horizontal since Ω is, and it is G-invariant since

$$\begin{aligned}(R_g)^*(f(\Omega)) &= f((R_g)^*\Omega)\\ &\overset{(a)}{=} f(\mathrm{Ad}(g^{-1})\Omega)\\ &\overset{(b)}{=} f(\Omega)\end{aligned}$$

where

(a) follows from Ω is G-equivariant.
(b) follows from f is G-invariant.

---

[5]Here we regard $\underbrace{\Omega \wedge \cdots \wedge \Omega}_{k \text{ times}}$ as a $\bigotimes_{i=1}^{k} \mathfrak{g}$-valued $2k$-form on P, so $f(\Omega)$ is well-defined.

To see that it is closed, direct computation shows

$$
\begin{aligned}
\mathrm{d}f(\Omega) &= f(\mathrm{d}\Omega\wedge\cdots\wedge\Omega)+\cdots+f(\Omega\wedge\cdots\wedge\mathrm{d}\Omega)\\
&\overset{(c)}{=} f(-\omega\wedge\Omega\wedge\cdots\wedge\Omega)+\cdots+f(\Omega\wedge\cdots\wedge-\omega\wedge\mathrm{d}\Omega)
\end{aligned}
$$

where (c) follows from the Bianchi identity. Since $\ker\omega$ is horizontal distribution, it suffices to show $\mathrm{d}f(\Omega)$ is horizontal to conclude $\mathrm{d}f(\Omega)=0$. Let X be a vertical vector field, by proof of Proposition 4.1.3 one has $\mathcal{L}_X f(\Omega)=0$ since $f(\Omega)$ is horizontal. Then by Cartan's formula one has

$$
\begin{aligned}
0 &= \mathcal{L}_X f(\Omega)\\
&= \mathrm{d}\circ\iota_X f(\Omega)+\iota_X\circ\mathrm{d}f(\Omega)\\
&= \iota_X\mathrm{d}f(\Omega)
\end{aligned}
$$

This completes the proof of (1).

For (2). The existence and uniqueness of $f(\Theta)$ follows from Lemma 7.2.1, and it is closed since

$$
\pi^*(\mathrm{d}f(\Theta)) = \mathrm{d}(\pi^*(f(\Theta))) = \mathrm{d}f(\Omega) = 0
$$

For (3). Suppose ω' is another connection on P. Let $P\times\mathbb{R}$ be a principal G-bundle over $M\times\mathbb{R}$, and $\widetilde{\omega}=(1-t)\omega+t\omega'$ is a connection on it with curvature $\widetilde{\Omega}$. Then $f(\widetilde{\Omega})$ gives a unique $2k$-form $\widetilde{\Theta}$ on $M\times\mathbb{R}$. If we use i_0 and i_1 to denote the maps from M to $M\times\{0\}$ and $M\times\{1\}$ respectively, then

$$
\begin{aligned}
f(\Theta) &= i_0^* f(\widetilde{\Theta})\\
f(\Theta') &= i_1^* f(\widetilde{\Theta})
\end{aligned}
$$

Since i_0 is homotopic to i_1, the homotopy invariance of de Rham cohomology implies $i_0^*, i_1^*\colon H^{2k}(M\times\mathbb{R},\mathbb{R})\to H^{2k}(M,\mathbb{R})$ coincide, and thus $[f(\Theta)]=[f(\Theta')]$. □

Theorem 7.2.1 (Chern-Weil homomorphism)**.** There is a ring homomorphism

$$
\begin{aligned}
W(P,\text{-})\colon I(\mathfrak{g}) &\to H^*(M,\mathbb{R})\\
f &\mapsto [f(\Theta)]
\end{aligned}
$$

Proof. For $f\in I^k(\mathfrak{g}), g\in I^l(\mathfrak{g})$, it suffices to show

$$
fg(\Theta) = f(\Theta)\wedge g(\Theta)
$$

Note that π^* is injective, so it suffices to check

$$
fg(\Omega) = f(\Omega)\wedge g(\Omega)
$$

which is clear. □

Remark 7.2.1.

7.3. **Transgression.** In this section we will show that for a given principal G-bundle P and a connection ω on it with curvature Ω, $[f(\Omega)] = 0 \in H^{2k}(P,\mathbb{R})$, where $f \in I^k(\mathfrak{g}), k \geq 1$. To see this, let us introduce the functorial Chern-Weil homomorphism. Given the following homomorphism between principal G-bundles

$$\begin{tikzcd} P' \arrow[r] \arrow[d, "\pi'"] & P \arrow[d, "\pi"] \\ M' \arrow[r, "\varphi"] & M \end{tikzcd}$$

where $P' = \varphi^* P$.

Proposition 7.3.1 (functoriality)**.** For all $f \in I(\mathfrak{g})$, we have

$$W(\varphi^* P, f) = \varphi^* W(P, f)$$

Proof. Given a connection $\omega \in \mathcal{A}(P)$ with curvature Ω, and let ω' denote the pullback connection $\widetilde{\varphi}^* \omega \in \mathcal{A}(P')$ with curvature Ω'. For any $f \in I(\mathfrak{g})$, it is clear

$$f(\Omega') = \widetilde{\varphi}^* f(\Omega)$$

Then

$$(\pi')^*(f(\Theta')) = \widetilde{\varphi}^* \pi^* f(\Theta) = (\pi')^* \varphi^* f(\Theta')$$

which implies $f(\Theta') = \varphi^* f(\Theta)$, since $(\pi')^*$ is injective. □

Example 7.3.1. Let $P = M \times G$ be the trivial principal G-bundle, consider

$$\begin{tikzcd} M \times G \arrow[r] \arrow[d, "\pi'"] & G \arrow[d, "\pi"] \\ M \arrow[r, "\varphi"] & \{\mathrm{pt}\} \end{tikzcd}$$

So for any $f \in I^k(\mathfrak{g}), k \geq 1$, we have

$$W(P, f) = \varphi^* W(G, f) = 0$$

since $W(G, f) \in H^{2k}(\{\mathrm{pt}\}) = 0$ if $k \geq 1$.

Remark 7.3.1. This example shows that if P is a trivial principal G-bundle, then the Chern-Weil homomorphism $W(P, \text{-})$ is trivial.

Now let us consider the following case

$$\begin{tikzcd} f^* P \arrow[r] \arrow[d, "\pi'"] & P \arrow[d, "\pi"] \\ P \arrow[r, "\varphi"] & M \end{tikzcd}$$

where $\varphi = \pi$. In fact we can write $f^* P$ down as

$$\begin{aligned} \varphi^* P &= \{(x', x) \in P \times P \mid \varphi(x') = \pi(x)\} \\ &= \{(x', x) \in P \times P \mid \pi(x') = \pi(x)\} \end{aligned}$$

It is clear that it has a global section, given by

$$\begin{aligned} s : P &\to \varphi^* P \\ x &\mapsto (x, x) \end{aligned}$$

so $\varphi^* P$ is a trivial principal bundle. Thus for any $f \in I^k(\mathfrak{g}), k \geq 1$, we have

$$W(\varphi^* P, f) = 0 \in H^{2k}(P)$$

However, functoriality implies

$$\begin{aligned} W(\varphi^* P, f) &= \varphi^* W(P, f) \\ &= \varphi^* [f(\Theta)] \\ &= \pi^* [f(\Theta)] \\ &= [f(\Omega)] \end{aligned}$$

This shows $[f(\Omega)] = 0$ in $H^{2k}(P, \mathbb{R})$.

8. Characteristic class

8.1. Chern class.

8.1.1. *Chern-Weil viewpoint.*

Proposition 8.1.1. Let $G = \mathrm{U}(n)$ with Lie algebra $\mathfrak{g} = \mathfrak{u}(n)$. For any $X \in \mathfrak{g}$, consider

$$\det(I - \frac{t}{2\pi\sqrt{-1}}X) = \sum_{k=0}^{n} c_k(X)t^k$$

Then

(1) For each $1 \leq k \leq n$, $c_k \in I(\mathfrak{g})$.
(2) $I(\mathfrak{g})$ is generated by $c_1, \ldots, c_n$.

Proof. For (1). For arbitrary $g \in G$, note that

$$\begin{aligned}\det(I - \frac{t}{2\pi\sqrt{-1}}\mathrm{Ad}(g)X) &= \det(I - \frac{t}{2\pi\sqrt{-1}}gXg^{-1}) \\ &= \det(I - \frac{t}{2\pi\sqrt{-1}}X)\end{aligned}$$

which implies $c_k \in I(\mathfrak{g})$.

For (2). Note that any $X \in \mathfrak{g}$ is diagonalizable, so without loss of generality we may assume $X = \mathrm{diag}\{\lambda_1, \ldots, \lambda_n\}$. Then $I(\mathfrak{g})$ consists of symmetric polynomials in $\lambda_1, \ldots, \lambda_n$. Then the proof follows since any symmetric polynomial can be expressed in terms of elementary symmetric functions and

$$\begin{aligned}c_1 &= -\frac{1}{2\pi\sqrt{-1}}(\lambda_1 + \cdots + \lambda_n) \\ &\vdots \\ c_n &= \left(-\frac{1}{2\pi\sqrt{-1}}\right)^n \lambda_1 \ldots \lambda_n.\end{aligned}$$

□

Let E be a complex vector bundle of rank n over M equipped with a Hermitian metric. Considering its frame bundle, we obtain a $\mathrm{U}(n)$-principal bundle $\pi: P \to M$, An arbitrary connection ω on P with curvature Ω gives a unique $2k$-form $c_k(\Theta)$ on M such that $\pi^*(c_k(\Theta)) = c_k(\Omega)$ by Chern-Weil theory.

Definition 8.1.1 (Chern class)**.** The k-th Chern class of E is defined as

$$c_k(E) := [c_k(\Theta)] \in H^{2k}(M, \mathbb{R}).$$

Definition 8.1.2 (Total Chern class)**.** The total Chern class of E is defined as $c(E) := \sum_{i=0}^{n} c_i(E)$, where $c_0(E) = 1$.

Definition 8.1.3 (Chern polynomial)**.** The Chern polynomial is defined as

$$c(t) = \det(I - \frac{t}{2\pi\sqrt{-1}}\Theta) = \sum_{k=0}^{n} c_k t^k$$

Proposition 8.1.2.

$$c_k \in H^{2k}(M,\mathbb{R})$$

Proof. Note that $\mathfrak{u}(n)$ consists of skew-Hermitian matrices, so for arbitrary $X \in \mathfrak{u}(n)$, one has

$$\begin{aligned}
\det(I - \frac{t}{2\pi\sqrt{-1}}X) &= \det(I + \frac{t}{2\pi\sqrt{-1}}\overline{X}^t) \\
&= \overline{\det(I - \frac{t}{2\pi\sqrt{-1}}X)} \\
&= \sum_{k=0}^{n} \overline{c}_k t^k
\end{aligned}$$

which implies $c_k = \overline{c}_k$. □

Proposition 8.1.3. Let $\overline{E}$ be the complex vector bundle obtained from E by taking the conjugate bundle. Then $c_k(\overline{E}) = (-1)^k c_k(E)$.

Proposition 8.1.4 (Whitney sum formula)**.** Let E,F be two complex vector bundles. Then

$$c(E \oplus F) = c(E)c(F)$$

Proof. If ∇^E, ∇^F are connections on E,F respectively with curvature Θ_E and Θ_F, then $\nabla^E \oplus \nabla^F$ gives a connection on $E \oplus F$ with curvature $\begin{pmatrix} \Theta_E & 0 \\ 0 & \Theta_F \end{pmatrix}$. This shows

$$\begin{aligned}
c(E \oplus F) &= \det\begin{pmatrix} I - \frac{1}{2\pi\sqrt{-1}}\Theta_E & 0 \\ 0 & I - \frac{1}{2\pi\sqrt{-1}}\Theta_F \end{pmatrix} \\
&= \det(I - \frac{1}{2\pi\sqrt{-1}}\Theta_E)\det(I - \frac{1}{2\pi\sqrt{-1}}\Theta_F) \\
&= c(E)c(F)
\end{aligned}$$

□

Corollary 8.1.1. Let E be a complex vector bundle and F be a trivial complex bundle. Then

$$c(E \oplus F) = c(E)$$

8.1.2. *Axiomatic viewpoint.* In this section we introduce the axiomatic definition of Chern classes which is given by Hirzebruch and Husemoller. To be explicit, the Chern classes of a complex vector bundle are given by the following four axioms.

(1) For each complex vector bundle E over a smooth manifold M and for each integer $i \geq 0$, the i-th Chern class $c_i(E) \in H^{2i}(M,\mathbb{R})$ is given and $c_0(E) = 1$.
(2) Let E be a complex vector bundle over M and let $f: M' \to M$ be a smooth map. Then
$$c(f^*E) = f^*(c(E)) \in H^*(M',\mathbb{R})$$
(3) Let E,F be two complex vector bundles. Then
$$c(E \oplus F) = c(E)c(F)$$

(4) $-c_1(\mathcal{O}_{\mathbb{CP}^1}(-1))$ is the generator[6] of $H^2(\mathbb{CP}^1,\mathbb{Z})$, where $\mathcal{O}_{\mathbb{CP}^1}(-1)$ is the tautological line bundle on $\mathbb{CP}^1$.

Theorem 8.1.1. Let E be a complex vector bundle. The Chern classes of E defined by the axioms are the same as those defined by Chern-Weil theory.

Proof. It suffices to verify the normalization axiom. Let $P = S^3 \to \mathbb{CP}^1$ be the Hopf fibration, with its principal U(1)-action

$$P \times \mathrm{U}(1) \to P$$
$$((z^0,z^1),\lambda) \mapsto (z^0\lambda, z^1\lambda)$$

With the convention used for associated bundles, the standard one-dimensional representation gives $\mathcal{O}_{\mathbb{CP}^1}(-1)$. Define a 1-form ω on P by

$$\omega = \frac{(\overline{z},\mathrm{d}z)}{(\overline{z},z)}$$

where $(\overline{z},\mathrm{d}z) = \overline{z}^0\mathrm{d}z^0 + \overline{z}^1\mathrm{d}z^1$ and $(\overline{z},z) = \overline{z}^0 z^0 + \overline{z}^1 z^1$. A straightforward verification yields that ω is a connection 1-form on P with curvature form

$$\Omega = \mathrm{d}\omega = \frac{(\overline{z},z)(\mathrm{d}\overline{z},\mathrm{d}z) - (z,\mathrm{d}\overline{z}) \wedge (\overline{z},\mathrm{d}z)}{(\overline{z},z)^2}$$

where

$$(\mathrm{d}\overline{z},\mathrm{d}z) = \mathrm{d}\overline{z}^0 \wedge \mathrm{d}z^0 + \mathrm{d}\overline{z}^1 \wedge \mathrm{d}z^1$$

Let U be the open subset of $\mathbb{CP}^1$ defined by $z^0 \neq 0$. Since $\mathbb{CP}^1 \setminus U$ is just a point, it suffices to compute $-\int_U c_1(\mathcal{O}_{\mathbb{CP}^1}(-1))$. If we set $w = z^1/z^0$, then w gives a local coordinate on U. Substituting $z^1 = z^0 w$ in the formula for Ω, one has

$$\Theta = \frac{\mathrm{d}\overline{w} \wedge \mathrm{d}w}{(1+w\overline{w})^2}$$

Thus

$$-\int_U c_1(\mathcal{O}_{\mathbb{CP}^1}(-1)) = -\int_U \frac{\sqrt{-1}}{2\pi}\frac{\mathrm{d}\overline{w} \wedge \mathrm{d}w}{(1+w\overline{w})^2} = 1$$

where the last equality can be derived from setting $w = re^{2\pi\sqrt{-1}\theta}$. □

8.1.3. *Chern character.* Let $G = \mathrm{U}(n)$ with Lie algebra $\mathfrak{g} = \mathfrak{u}(n)$ and let E be a complex vector bundle over a smooth manifold M with frame bundle P. Consider the following power series which is Ad-invariant.

$$\mathrm{tr}\left(\exp\left(\frac{-1}{2\pi\sqrt{-1}}X\right)\right), \qquad X \in \mathfrak{u}(n).$$

Suppose ω is a connection 1-form on P with curvature Ω. Then by Chern-Weil theory

$$\mathrm{tr}\left(\exp(\frac{-1}{2\pi\sqrt{-1}}\Omega)\right)$$

[6]In other words, $c_1(\mathcal{O}_{\mathbb{CP}^1}(-1))$ evaluated on the fundamental 2-cycle $\mathbb{CP}^1$ equals -1, which is equivalent to saying $\int_{\mathbb{CP}^1} c_1(\mathcal{O}_{\mathbb{CP}^1}(-1)) = -1$.

descends to an element in $H^*(M,\mathbb{R})$, which is denoted by $\mathrm{ch}(E)$ and called the Chern character of E.

Proposition 8.1.5. If $c(E) = \prod_{i=1}^n (1+x_i)$, then $\mathrm{ch}(E) = \sum_{i=1}^n \exp x_i$.

Proof. For arbitrary $X \in \mathfrak{g}$, without loss of generality we can assume $X = \mathrm{diag}\{\sqrt{-1}\lambda_1, \dots, \sqrt{-1}\lambda_n\}$. Then

$$\mathrm{tr}\left(\exp(\frac{-1}{2\pi\sqrt{-1}}X)\right) = \sum_{i=1}^n \exp(-\frac{\lambda_i}{2\pi})$$

On the other hand, one has

$$\det(I - \frac{1}{2\pi\sqrt{-1}}X) = (1-\frac{\lambda_1}{2\pi})\dots(1-\frac{\lambda_n}{2\pi})$$

□

Remark 8.1.1 (splitting principle). It is interesting to note that $c(E) = \prod_i c(L_i)$, where L_i is a line bundle with curvature given by $-2\pi\sqrt{-1}x_i$. Thus, as far as the Chern classes are concerned, the vector bundle E behaves as a direct sum of the line bundles $L_1 \oplus \cdots \oplus L_n$. This phenomenon is called the splitting principle. Using this notation, the Todd classes are defined by

$$\mathrm{td}(E) = \prod_i \frac{x_i}{1-e^{-x_i}}$$

The L-genus is defined by

$$L(E) = \prod_i \frac{x_i}{\tanh x_i}$$

and the $\widehat{A}$-genus is defined by

$$\widehat{A}(E) = \prod_i \frac{x_i/2}{\sinh(x_i/2)}$$

Proposition 8.1.6. Suppose E_1 and E_2 are two complex vector bundles. Then

$$\mathrm{ch}(E_1 \oplus E_2) = \mathrm{ch}(E_1) + \mathrm{ch}(E_2)$$
$$\mathrm{ch}(E_1 \otimes E_2) = \mathrm{ch}(E_1) \wedge \mathrm{ch}(E_2)$$

8.1.4. *Chern classes of complex projective space.*

Definition 8.1.4 (Chern class of complex manifold)**.** Let X be a complex manifold. Then its k-th Chern class $c_k(X)$ is defined to be the Chern class of its holomorphic tangent bundle.

Theorem 8.1.2. The total Chern class of $\mathbb{CP}^n$ is equal to $(1+\omega)^{n+1}$, where ω is a suitably chosen generator for $H^2(\mathbb{CP}^n,\mathbb{Z})$.

8.2. **Pontryagin class.** Let E be a real vector bundle of rank n over a smooth manifold M. Then its frame bundle P be a principal O(n)-bundle. For any $X \in \mathfrak{o}(n)$, consider

$$\det(I - \frac{t}{2\pi}X) = \sum_{k=0}^{n} q_k(X)t^k$$

By the same argument one can show $q_k \in I(\mathfrak{g})$, so if we pick an arbitrary connection ω on P with curvature Ω, then it gives rise to a closed $2k$-form $q_k(\Theta)$ on M for each $k \in \mathbb{Z}_{\geq 0}$. Since $X + X^t = 0$, one has

$$\det(I + \frac{t}{2\pi}X) = \det(I - \frac{-t}{2\pi}X)$$

which implies

$$q_k(X) = q_k(-X) = (-1)^k q_k(X)$$

Thus we can conclude $q_k = 0$ when k is odd.

Definition 8.2.1 (Pontryagin class)**.** The k-th Pontryagin class of E is defined as $[p_k(\Theta)] := [q_{2k}(\Theta)] \in H^{4k}(M, \mathbb{R})$, where $0 \leq k \leq \lfloor n/2 \rfloor$.

Definition 8.2.2 (Pontryagin class of manifold)**.** Let M be a smooth manifold. Then its k-th Pontryagin class $p_k(M)$ is defined to be the Pontryagin class of its tangent bundle.

Proposition 8.2.1 (Whitney sum formula)**.** Let E, F be two vector bundles. Then

$$p(E \oplus F) = p(E)p(F)$$

Corollary 8.2.1. Let E be a vector bundle and F be a trivial bundle. Then

$$p(E \oplus F) = p(E)$$

Corollary 8.2.2. $p(S^n) = 1$.

Proof. Note that $S^n \hookrightarrow \mathbb{R}^{n+1}$. If NS^n denotes the normal bundle of S^n, then $TS^n \oplus NS^n$ is a trivial bundle, which implies $p(S^n) = p(TS^n) = 1$. □

Proposition 8.2.2. Let E be a vector bundle with its complexification $E_{\mathbb{C}} = E \otimes \mathbb{C}$. Then

$$p_k(E) = (-1)^k c_{2k}(E_{\mathbb{C}})$$

Proof. See Theorem 4.1 in Chapter 12 of [KN96]. □

Corollary 8.2.3. Let E be a complex vector bundle and let $E_{\mathbb{R}}$ be its underlying real bundle. Then

$$1 - p_1(E_{\mathbb{R}}) + p_2(E_{\mathbb{R}}) + \cdots \pm p_n(E_{\mathbb{R}}) = (1 - c_1(E) + c_2(E) + \cdots \pm c_n(E))(1 + c_1(E) + c_2(E) + \cdots + c_n(E))$$

In particular, if X is a complex manifold, then $p_1(X) = c_1(X)^2 - 2c_2(X)$.

Example 8.2.1. Since by Theorem 8.1.2 the total Chern class $c(\mathbb{CP}^n)$ equals $(1+\omega)^{n+1}$, where ω is a suitably chosen generator of $H^2(\mathbb{CP}^n, \mathbb{Z})$. Then

$$\begin{aligned}(1 - p_1(\mathbb{CP}^n) + p_2(\mathbb{CP}^n) - \dots) &= (1-\omega)^{n+1}(1+\omega)^{n+1} \\ &= (1-\omega^2)^{n+1}\end{aligned}$$

Therefore the total Pontryagin class $p(\mathbb{CP}^n) = (1+\omega^2)^{n+1}$. In other words, one has

$$p_k(\mathbb{CP}^n) = \binom{n+1}{k}\omega^{2k}$$

Corollary 8.2.4. $\mathbb{RP}^2$ cannot be embedded into $\mathbb{R}^3$.

8.3. **Euler class.** Let $n = 2m$ and consider Lie group $\mathrm{SO}(n)$ with Lie algebra $\mathfrak{so}(n)$. A basic fact is that $\mathfrak{so}(n) = \mathfrak{o}(n)$ since $\mathrm{SO}(n)$ is the identity component of $\mathrm{O}(n)$, and thus $\mathfrak{so}(n)$ is the set of all skew-symmetric matrices. For $A = (a_{i,j}) \in \mathfrak{so}(n)$, the Pfaffian of A is defined as

$$\mathrm{Pf}(A) := \frac{1}{2^m m!}\sum_{\sigma\in S_n}\mathrm{sign}(\sigma)\prod_{i=1}^m a_{\sigma(2i-1),\sigma(2i)}$$

A direct computation shows $\mathrm{Pf} \in I^m(\mathfrak{so}(n))$, and thus it gives a characteristic class by Chern-Weil theory.

Definition 8.3.1 (Euler class)**.** Let E be an oriented vector bundle of rank $2m$. Then $[\frac{1}{(2\pi)^m}\mathrm{Pf}(\Theta)] \in H^{2m}(M,\mathbb{R})$ is called the Euler class of E, which is denoted by $e(E)$.

Proposition 8.3.1. Let A,B be two skew-symmetric matrices. Then

$$\mathrm{Pf}(A\oplus B) = \mathrm{Pf}(A)\mathrm{Pf}(B)$$

Corollary 8.3.1 (Whitney sum formula)**.** Let E,F be two oriented vector bundles. Then

$$e(E\oplus F) = e(E)e(F)$$

Proposition 8.3.2. Let E be an oriented vector bundle of rank $2m$. Then

$$c_{2m}(E\otimes\mathbb{C}) = e((E\otimes\mathbb{C})_{\mathbb{R}})$$

As a consequence, if E is a complex vector bundle of rank n, then $c_n(E) = e(E_{\mathbb{R}})$.

Corollary 8.3.2. Let E be an oriented vector bundle of rank $2m$. Then $p_m(E)$ equals the square of the Euler class $e(E)$.

Theorem 8.3.1 (Gauss-Bonnet-Chern)**.** Let M be a compact oriented $2m$-dimensional manifold. Then

$$\int_M e(TM) = \chi(M)$$

Corollary 8.3.3 (Gauss-Bonnet-Chern)**.** Let X be a compact complex n-dimensional manifold. Then

$$\int_X c_n(X) = \chi(X).$$

9. The classifying space

In the last section, we have defined characteristic classes in a geometric viewpoint. However, they are topological invariants. In this section, we work in the category of topological spaces (in particular, CW-complexes) instead of smooth manifolds, and give another explanation about characteristic classes.

9.1. **The universal G-bundle.**

Definition 9.1.1 (contractible)**.** A topological space is called contractible, if it is homotopy equivalent to a point.

Definition 9.1.2 (weak homotopy)**.** Let X,Y be topological spaces. X is said to be weak homotopy equivalent to Y, if there exists a continuous map $f: X \to Y$ such that f induces isomorphisms between homotopy groups of X and Y.

Definition 9.1.3 (weakly contractible)**.** A topological space X is called weakly contractible, if it is weakly homotopy equivalent to a point.

Example 9.1.1. A contractible space is weakly contractible.

Proposition 9.1.1. For any topological space, there exists a CW-complex which is weakly homotopy equivalent to it.

Theorem 9.1.1 (Whitehead)**.** A weak homotopy equivalence between CW-complexes is the same as homotopy equivalence.

Corollary 9.1.1. A CW-complex is weakly contractible if and only if it is contractible.

Definition 9.1.4 (classifying space)**.** Let $EG \to BG$ be a principal G-bundle, where EG,BG are topological spaces. If EG is weakly contractible, then

(1) BG is called a classifying space for G.
(2) EG is called a universal G-bundle.

Proposition 9.1.2. If the a classifying space for G exists, then there exists a classifying space $EG \to BG$ for G such that BG is CW-complex.

Proof. Suppose $P \to B$ is a classifying space for G, where P,B are topological spaces. By Proposition 9.1.1, there exists a CW-complex BG and a weak homotopy equivalence $f: BG \to B$. By exact homotopy sequence of fibration, one has

$$\begin{tikzcd}
\cdots \to \pi_{n+1}(B) \arrow[r] \arrow[d, "\cong"] & \pi_n(G) \arrow[r] \arrow[d, "\cong"] & \pi_n(P) \arrow[r] \arrow[d] & \pi_n(B) \arrow[r] \arrow[d, "\cong"] & \pi_{n-1}(G) \to \dots \arrow[d, "\cong"] \\
\cdots \to \pi_{n+1}(EG) \arrow[r] & \pi_n(G) \arrow[r] & \pi_n(f^*P) \arrow[r] & \pi_n(EB) \arrow[r] & \pi_{n-1}(G) \to \dots
\end{tikzcd}$$

Then $f^*P \to BG$ is a classifying space for G by the five-lemma. □

Theorem 9.1.2. Let $EG \to BG$ be a universal G-bundle. For all CW-complexes X, the following map is bijective:

$$\begin{aligned} \Phi\colon [X,BG] &\to \mathcal{P}_G X \\ f &\mapsto f^*P \end{aligned}$$

where $[X,BG]$ denotes the set of all continuous maps up to homotopy.

Proof. See [Mit01]. □

Remark 9.1.1. This theorem explains why BG is called classifying space, since it can be used to classify principal G-bundles over a given CW-complex.

Theorem 9.1.3. For any topological group G, a classifying space for G exists, and it is unique up to G-homotopy equivalence.

Proof. See [Mil56]. □

Proposition 9.1.3. For a discrete group G, $PK(G,1) \to K(G,1)$ is the universal G-bundle, and hence $K(G,1)$ is a classifying space for G.

Proof. It is clear that the path space $PK(G,1)$ is contractible. □

Remark 9.1.2. In [Liu22] we have already computed $K(G,1)$ for groups, for example, $K(\mathbb{Z},1) = \mathbb{S}^1$, $K(\mathbb{Z}_2,1) = \mathbb{RP}^\infty$ and so on.

Proposition 9.1.4. $V_n(\mathbb{R}^\infty) \to Gr_n(\mathbb{R}^\infty)$ is a universal $\mathrm{GL}(n,\mathbb{R})$-bundle, and hence $Gr_n(\mathbb{R}^\infty)$ is a classifying space for $\mathrm{GL}(n,\mathbb{R})$.

Proof. It suffices to show $V_n(\mathbb{R}^\infty)$ is contractible. Since we have already computed low-dimensional homotopy groups of $V_n(\mathbb{R}^N)$ in [Liu22], and then the telescope construction completes the proof. □

Corollary 9.1.2. For all CW-complexes X, $[X, Gr_n(\mathbb{R}^\infty)] \to \mathrm{Vect}_n^{\mathbb{R}} X$ is bijective.

Proof. See Remark 2.3.2. □

Remark 9.1.3. The analogous result also holds with $\mathbb{R}$ replaced by $\mathbb{C}$.

9.2. **homotopical properties of classifying spaces.** In this section we collect some homotopical properties of classifying spaces.

Theorem 9.2.1. For any topological group G, G is weakly equivalent to the loop space $\Omega(BG)$.

Corollary 9.2.1. For $n \geq 1$, $\pi_n(BG) = \pi_{n-1}(G)$.

Theorem 9.2.2. Let G be a topological space and H a subgroup. Then the homotopy fiber of $BH \to BG$ is G/H, up to weak equivalence.

Theorem 9.2.3. Let G be a topological space and H a subgroup. Then there is a fibration $BH \to BG \to B(G/H)$.

Example 9.2.1. The exact sequences $1 \to \mathrm{SO}(n) \to \mathrm{O}(n) \to \mathbb{Z}_2 \to 1$ and $1 \to \mathrm{SU}(n) \to \mathrm{U}(n) \to S^1 \to 1$ give rise to fibration

$$B\,\mathrm{SO}(n) \to B\mathrm{O}(n) \to \mathbb{RP}^\infty$$

and

$$B\,\mathrm{SU}(n) \to B\,\mathrm{U}(n) \to \mathbb{CP}^\infty$$

9.3. **Another viewpoint on characteristic classes.**

Proposition 9.3.1. The cohomology ring of $B\,\mathrm{U}(n)$ with integer coefficients is $\mathbb{Z}[c_1,\ldots,c_n]$.

Proof. If we consider $\mathrm{U}(n-1)$ as a subgroup of $\mathrm{U}(n)$, then we have the following filtration

$$\begin{tikzcd} S^{2n-1} \cong \mathrm{U}(n)/\mathrm{U}(n-1) \arrow[r] & B\,\mathrm{U}(n-1) \arrow[d] \\ & B\,\mathrm{U}(n) \end{tikzcd}$$

Apply the Leray spectral sequence to this fibration and use the fact that the cohomology ring of $B\,\mathrm{U}(1) = \mathbb{CP}^\infty$ is $\mathbb{Z}[c_1]$ to conclude. $\square$

Definition 9.3.1 (universal Chern class)**.** The generators $c_1, \cdots, c_n$ of $H^*(B\,\mathrm{U}(n), \mathbb{Z})$ are called the universal Chern classes of $\mathrm{U}(n)$-bundles.

Definition 9.3.2 (Chern class)**.** The k-th Chern class of the $\mathrm{U}(n)$-bundle $\pi\colon E \to M$ with classifying map $f_\pi\colon M \to B\,\mathrm{U}(n)$ is defined as

$$c_k(E) := f_\pi^*(c_k) \in H^{2k}(M, \mathbb{Z})$$

Proposition 9.3.2. The cohomology ring of $B\mathrm{O}(n)$ with $\mathbb{Z}_2$ coefficients is $\mathbb{Z}_2[w_1,\ldots,w_n]$.

Proof. The proof is the same as above; just note that the cohomology ring of $\mathbb{RP}^\infty$ with $\mathbb{Z}_2$ coefficients is $\mathbb{Z}_2[w_1]$. $\square$

Definition 9.3.3 (universal Stiefel-Whitney class)**.** The generators $w_1, \cdots, w_n$ of $H^*(B\mathrm{O}(n), \mathbb{Z}_2)$ are called the universal Stiefel-Whitney classes of $\mathrm{O}(n)$-bundles.

Definition 9.3.4 (Stiefel-Whitney class)**.** The k-th Stiefel-Whitney class of the $\mathrm{O}(n)$-bundle $\pi\colon E \to M$ with classifying map $f_\pi\colon M \to B\mathrm{O}(n)$ is defined as

$$w_k(E) := f_\pi^*(w_k) \in H^k(M, \mathbb{Z}_2)$$

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